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Master's Thesis

Precise, Grade-Specific Text Simplification via Weighted Adapter Merger for Second Language Acquisition

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Declaration of Authorship

Declaration

I hereby confirm that the thesis presented here is my own work, with all assistance acknowledged. I assure that the electronic version is identical in content to the printed version of the Master's thesis.

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Abstract

Precise, Grade-Specific Text Simplification via Weighted Adapter Merger for Second Language Acquisition

by William LaCroix

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This thesis evaluates whether modular adapter composition can enable controllable readability in neural sentence simplification for second language learners. Educational reading materials must align with learner proficiency, yet most simplification systems treat complexity reduction as a binary task rather than targeting specific difficulty levels. To explore an alternative approach, this study trains multiple low-rank adapters on sentence simplification data labeled by readability grade level. These grade-specific adapters are then combined via weighted parameter merging in an attempt to approximate intermediate difficulty levels without training a separate model for every possible target grade. Experiments are conducted on a preprocessed version of the WikiLarge dataset annotated with Flesch-Kincaid grade level labels. Results show that grade-specific adapters capture meaningful differences in linguistic complexity but only approximate their intended readability targets. Even the strongest models remain roughly one grade level away from the target on average. Adapter merging allows approximate interpolation between adjacent levels but does not outperform the best single-adapter fine-tuned models and produces unstable behavior under many merge configurations. Overall, the findings show that adapter merging enables modular recombination of simplification models but does not provide the precise readability control required for inter-grade alignment in text simplification for language learners.

Keywords: text simplification, second language education, readability control, controllable text generation, adapter merging, parameter-efficient fine-tuning, evaluation

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For those who come after

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ACRONYMS

ΔFKGL change in Flesch-Kincaid Grade Level

BERTScore F1 Bidirectional Encoder Representations from Transformers F1 Score

BNB Bernoulli Naïve Bayes

BTM Branch-Train Merge

BTX Branch-Train Mix

CAT concatenation

CEFR Common European Framework of Reference for Languages

CL controlled language

DARE Domain-Aware Representation Enhancement

DARE_TIES Domain-Aware Representation Enhancement with Task-Informed Embedding
Similarity

FKGL Flesch-Kincaid Grade Level

HF Hugging Face

L2 second language

LLaMA3 LLaMA3.2-3B-Instruct

LLM large language model

LoRA Low-Rank Adapter

ML machine learning

MT machine translation

NLP natural language processing

NMT neural machine translation

NTS neural text simplification

OTS off-the-shelf

PEFT Parameter-Efficient Fine-Tuning

RNN recurrent neural network

RQ research question

RQ1 Research Question 1

RQ2 Research Question 2

RQ3 Research Question 3

SARI System output Against References and against the Input

seq2seq sequence-to-sequence

SLA second language acquisition

SMT statistical machine translation

STS statistical text simplification

SVD Singular Value Decomposition

TIES Task-Informed Embedding Similarity

TS text simplification

w@ weighting

ws@ window-size

ZPD Zone of Proximal Development

CHAPTER 1

INTRODUCTION

```
1 def simplify_text(source_text: str, difficulty_level: int) -> str:
2     """Transform a source text into a version aligned with a target
3     proficiency level.
4
5     Args:
6         source_text (str): Input text of arbitrary complexity.
7         target_level (int): Desired readability level (e.g., grade 5, grade 6).
8
9     Returns:
10         str: A text aligned with target_level while preserving meaning.
11     """
```

The seemingly simple act of transforming a complex original text into a simpler, more understandable form is known as text simplification (TS). This process has applications across myriad domains, from education to accessibility, technical documentation, and public information, among many others. For this thesis, we will limit the discussion of TS to the domain of foreign language education, which we will refer to from here on out as second language acquisition (SLA). First, we must make a distinction between text *simplification* and text *summarization*: both simplification and summarization tasks often involve reducing text length and replacing obscure vocabulary, but text summarization focuses on preserving the *content* of the message without regard to language complexity, while text simplification focuses primarily on transforming the *structure* of the text into a more readable form.

Within educational contexts, readability and text complexity are key features since, broadly speaking, second language (L2) learning materials are separated by proficiency level or grade (e.g., A1-C2 on the Common European Framework of Reference for Languages (CEFR), grades 1-12 of the American school system, etc.) by their complexity. At the same time, access to appropriately simplified texts for language practice is generally limited to the specific curriculum with which

the learner is engaged. Outside of this narrow context, real-world simplified content is often confined to texts written in “normal” or “simple” language (e.g., Simple English Wikipedia), with little regard for the vast spectrum of abilities a language learner will experience on their learning journey. Rather than a single estimation of “simple”, an idealized scenario would be content that is tailored to the specific ability of a given student. Although there is a growing industry surrounding the production of target-level reading materials (e.g., graded readers), these materials are limited in availability and variety, while still suffering from a steep increase in complexity learners face when transitioning between levels, for example: A2 to B1, grade 5 to grade 6. In order to provide level-appropriate content across this spectrum of language abilities, there needs to be a system capable of scaling source texts to any given proficiency, with precise control over the difficulty of the generated text. Thus, this research aims to build a TS model capable of outputting text targeting smaller specific increments between levels (e.g., grade 4.2, grade 6.9, etc.) to address the lack of content targeted along the spectrum of language ability. The remainder of this chapter will provide an introduction to the pedagogical basis behind simplified texts for SLA, previous methods of text simplification, the shortcomings of current techniques, the aims, specific objectives, and considerations of this research, the significance of this work, and finally the potential limitations of the current research.

1.1 Background

1.1.1 Language Input and The Input Hypothesis

Regardless of the setting, be it classroom learning or self-study, the role of input (i.e., the language that learners are exposed to through listening or reading) in language acquisition cannot be understated. Indeed, there exists an entire lineage of language learning pedagogy centered around input-only methods, based on the work of Stephen Krashen and the so-called “input hypothesis”, one of the five hypotheses of Krashen’s Monitor Model of SLA. This hypothesis consists of four components, as laid out in pages 22-23 of Krashen (1982), *Principles and Practice in Second Language Acquisition*:

1. The distinction between, and focus on, “acquisition”, as opposed to “learning”.
2. Language acquisition involves understanding language just beyond the learner’s current ability. This will be referred to as $i+1$ from here on, denoting an incremental increase over the current ability.
3. Input must contain $i+1$ to be useful for acquisition, though it should not be limited to $i+1$. Given enough comprehensible input, $i+1$ will be provided.
4. Language production ability is both naturally and gradually emerging, not taught.

A related concept is Vygotsky's Zone of Proximal Development (ZPD), a foundational idea in educational psychology that describes the range of tasks a learner cannot yet perform independently but can accomplish with guidance (Vygotsky, 1978). The ZPD emphasizes instructional scaffolding, where a more knowledgeable individual provides support to help the learner master tasks that lie just beyond their current competence. While both the ZPD and Krashen's $i+1$ highlight the importance of exposure to material beyond the learner's present level, they differ in how they frame the learner's role. Krashen's $i+1$ encourages learners to engage with input slightly beyond their current level on their own, allowing room for ambiguity and incidental acquisition (Krashen, 1982). In contrast, the ZPD framework assumes that the learner requires assistance to complete the task and that instructional support is a necessary part of the learning process. If $i+1$ represents content at the boundary of comprehension, then ZPD can be seen as $i+n$, where n is large enough to require guidance but still within the learner's potential reach. This comparison illustrates the shared focus on incremental challenges in both models while clarifying their pedagogical differences. Both approaches highlight the critical role of appropriately leveled input in fostering language development, especially when learners are consistently exposed to material that is engaging yet slightly beyond their current proficiency.

Extensive reading, that is, reading for enjoyment as opposed to reading specifically to learn, is a popular input-focused method of language practice (Bamford and Day, 2004). Yet given that Google Books Ngram Corpus (2010) estimates that there are over 130 million books in existence, and even the most voracious reader might only manage around 4,500 books in a lifetime, the sheer scale of available reading material can be overwhelming. This raises a practical question for learners: where should they begin? More often than not, children's books are paradoxically too complex while also being too boring, as they often present both linguistic and cultural challenges to L2 readers and may not engage adult interests (Waring 2009, as cited in Shaffer 2012; Hitosugi and Day 2004). Young adult literature (only partially) alleviates the boredom while increasing the complexity, and adult native speaker materials are by-and-large too complex to be used as extensive reading materials (Bamford and Day, 2004, Hitosugi and Day, 2004). Therefore, appropriate texts for extensive reading need to balance challenge and engagement: the text should be interesting enough that a learner stays motivated while still being at a level appropriate for comprehension, that is, $i+1$ (Liu and Zhang, 2018). Learner motivation and engagement are also crucial to the success of input-based methods like extensive reading. According to Illeris's theory of learning, engagement depends not only on cognitive challenges but also on emotional and social dimensions, including interest, relevance, and learner identity (Illeris, 2003). Similarly, Mori (2002) emphasizes that motivation in foreign language reading is shaped by a learner's intrinsic interest, perceived relevance, and confidence in their reading ability. Therefore, for reading practice to be effective, materials should not only match the learner's proficiency level but also align with their personal interests and goals. This alignment supports sustained engagement and deepens the learning process across all dimensions.

1.1.2 Graded Readers

One approach that aims to satisfy both linguistic accessibility and learner engagement is the use of graded readers. Credited to Dr. Michael West in the 1920s, though they would not come into vogue until the publication of Jon Milne’s Heinemann Graded Readers series (Milne, 1977), the concept behind graded readers is simple: adapted works or original writing targeted to a specific proficiency level to provide extensive reading materials for language learners. These readers are often written with a limited set of vocabulary and grammar constructions, aligned with L2 pedagogical frameworks; for example: A1 readers contain roughly 300 unique headwords and simple grammatical structures (Rodrigo, 2016). While this style of writing does a better job of tailoring input to language learners’ needs, materials still suffer from large increases in complexity between levels (e.g., the distance between A2 and B1). This approach makes these materials more broadly suited to students at a particular level *in general*, but fails to address the individual learner’s needs and abilities.

While graded readers offer level-appropriate materials for language learners, they are typically constructed around discrete, pre-defined proficiency levels (e.g., CEFR A1, A2, B1, etc.), each encompassing a relatively wide range of linguistic complexity (Rodrigo, 2016). Consequently, learners often experience noticeable jumps in text difficulty when transitioning between levels, as the range of vocabulary, syntactic structures, and discourse patterns can increase substantially from one band to the next. This “stair-step effect” can result in learner frustration, particularly for those at the margins of a given level, who may find texts either too simple to be challenging or too complex to be comprehensible (Nation, 2001, Waring, 2009). In addition, graded readers typically assume an average or general learner profile for each level, without accommodating individual variation in vocabulary size, syntactic parsing ability, or prior topic familiarity (Crossley et al., 2008). As a result, the one-size-fits-all nature of these materials limits their ability to support personalized language development. Precise, graded TS offers a solution to this issue by enabling dynamic generation of texts at smaller incremental proficiency steps (e.g., grade 4.2, grade 4.3), more closely matching a learner’s current linguistic ability and zone of proximal development (Vygotsky, 1978), and thereby supporting continuous, sustained language growth. Therefore, one of the primary goals of this research is to address this shortcoming of current graded texts, which is further discussed in Section 1.1.4 - The Research Problem.

1.1.3 Readability in Text Simplification

In the context of TS for L2 learners, readability refers to the ease with which a text can be processed and understood by the reader and is assessed using quantifiable metrics. While early readability formulae such as Flesch-Kincaid or Gunning Fog Index rely on surface-level features like sentence length and syllable count, contemporary research has expanded these

measures to include more granular linguistic and cognitive variables such as syntactic dependency depth, lexical frequency distributions, and psycholinguistic features like age of acquisition or concreteness ratings (Alva-Manchego et al., 2021, Siddharthan, 2014). However, this study’s primary concern is the former: transparent readability scores as a means of evaluating simplified texts. Modern TS systems often use these traditional readability metrics to approximate the complexity of output texts, providing an automated method for evaluating simplification models.

Although such metrics have well-known limitations, for example their inability to capture deeper semantic or discourse-level complexity, they remain useful for benchmarking simplification performance, particularly when paired with sentence-level simplification tasks common in data-driven approaches (Alva-Manchego et al., 2021). For these reasons, the present study adopts a model-agnostic evaluation approach centered on these established readability metrics, allowing for systematic comparisons of simplification output across models and settings without requiring psycholinguistic annotations or user studies (Agrawal and Carpuat, 2023). By constraining the focus to these metrics, this work aligns with practical needs in L2 educational contexts, where the goal is often to quickly assess the suitability of texts for learners with limited proficiency, rather than to model their cognitive processing in complete detail (Crossley et al., 2008).

1.1.4 Research Problem

Accurate control over output readability is an important goal in TS for SLA, where pedagogical input should closely match the learner’s proficiency. Recent advances in neural text generation have enabled simplification systems to produce fluent and semantically faithful outputs, yet most remain limited in their ability to flexibly modulate complexity. Prior approaches have attempted to control simplification through explicit linguistic targets, control tokens, or prompt-based conditioning. While these methods have been effective, they often lack precision in mapping prompts to concrete readability levels and are difficult to adapt smoothly across a continuum of learner proficiency. This thesis explores an alternative architectural strategy: using weighted mixtures of fine-tuned adapters trained on texts of varying complexity to enable approximate control over output readability levels. This method sidesteps the need for prompt calibration and mitigates the impracticality of assembling training data for every incremental difficulty level by spreading interpolation across a limited set of fixed readability levels.

1.2 Research Questions

The following research questions (RQs) guide this investigation:

Research Question 1: Do sentence simplification adapters trained on discrete grade levels reliably generate output that corresponds to their target grade level, as measured by standard readability metrics?

This question serves to establish the baseline effectiveness of using adapter modules trained on grade-specific data. It evaluates whether discrete adapters can reliably produce outputs aligned with their intended complexity levels, forming the foundation for more advanced composition strategies.

Research Question 2: Can combining grade-specific adapters through weighted parameter merging improve the system’s ability to generate text at intended readability levels?

This question evaluates whether merging adapter modules leads to more consistent and controllable simplification outcomes compared to using single adapters. The goal is to determine if weighted composition can reduce variability and better align outputs with target readability levels, thereby improving the precision of the simplification process.

Research Question 3: Can a learnable weighting scheme over grade-specific adapters enable the generation of text at arbitrary intermediate readability levels not explicitly represented in training data?

This question investigates whether it is possible to interpolate between discrete simplification levels by adjusting adapter weights, effectively allowing the system to “turn a dial” to target any point along the readability spectrum. Since parallel corpora for every potential grade level are impractical to obtain, this approach offers a method for precisely controlling the generation of inter-level simplification through model composition at runtime.

Together, these questions support a broader investigation into the feasibility of adapter-based model composition for controllable TS. The thesis assesses whether this architecture can enable precise control over output readability compared to existing approaches. By grounding the experiments in both fixed-level performance and inter-grade interpolation, we seek to advance methods that better align simplification output with the diverse and continuous proficiency range of L2 learners.

1.3 Contributions

This thesis investigates whether modular adapter composition can provide controllable readability in neural text simplification (NTS) for L2 learners. Specifically, it evaluates whether grade-specific fine-tuned adapters can be combined through (weighted) parameter merging to approximate intermediate levels of linguistic complexity without requiring separate models trained for every target grade.

The main contributions of this work are as follows:

- **An empirical evaluation of adapter-based readability control for TS.** This thesis investigates whether sentence simplification adapters trained on discrete readability levels can reliably generate outputs aligned with their intended grade targets.
- **A systematic analysis of adapter merging for controllable text simplification.** Multiple merging strategies, weighting schemes, and merge configurations are evaluated to determine whether combining grade-specific adapters improves readability control or enables interpolation between trained difficulty levels.
- **An investigation of readability-driven training for simplification models.** This thesis examines the effects of using readability grade level as the primary optimization target when training grade-specific adapters for simplification, evaluating how directly optimizing readability influences output complexity and model behavior.
- **Empirical evidence on the limits of merged interpolation for readability control.** The results show that while adapter merging allows approximate interpolation between adjacent readability levels, it does not substantially improve precision beyond individually fine-tuned adapters.
- **Implications for L2-oriented text simplification.** By examining the relationship between model composition, readability metrics, and pedagogical objectives, the thesis clarifies both the potential and limitations of current approaches to controllable TS for SLA applications.

1.4 Thesis Scope and Boundaries

This thesis implements a controllable text simplification system based on the LLaMA3.2-3B-Instruct (LLaMA3) model fine-tuned using Low-Rank Adapter (LoRA). Training data derives from the WikiLarge corpus, with additional pre-processing and Flesch-Kincaid Grade Level (FKGL)-based grade annotation. The models perform grade-conditioned generation, where

control is expressed through FKGL-derived grade labels (Grades 2-12). These models are then combined using (weighted) adapter merging for steering output text complexity. In addition to grade-level evaluation, we evaluate model outputs using automatic simplification metrics, including System output Against References and against the Input (SARI), Bidirectional Encoder Representations from Transformers F1 Score (BERTScore F1), and perplexity.

The scope of this work is intentionally limited in several respects.

First, due to the nature of the dataset used, all experiments are conducted exclusively in English. Therefore, the findings do not necessarily generalize to other languages, multilingual, or cross-lingual simplification settings.

Second, the simplification system operates only at the sentence level. We do not attempt to model document-level structure, discourse coherence, or inter-sentential dependencies. Each sentence is simplified independently, again due to the structure of our dataset.

Third, readability control relies exclusively on FKGL-based grade measurement. We do not incorporate any alternative readability metrics, curriculum-aligned standards, or human-validated proficiency scales.

Finally, evaluation uses only automatic measurements. As we do not conduct human comprehension studies, conclusions about pedagogical effectiveness are limited to metric-based analysis.

These boundaries help define the methodological and empirical limits of the present research. The results should therefore be interpreted as evidence regarding sentence-level, FKGL-aligned simplification within an English-only, automatic evaluation framework.

1.5 Thesis Structure

This introduction provides the context and motivation for the research and establishes the framework used to evaluate our approach. It also provides considerations for the limitations of the current research.

Chapter 2 reviews prior literature and research into methods for automated TS, including controlled simplification, model merging, and readability evaluation metrics, as well as situating this study within the broader discourse on L2 pedagogy and TS.

Chapter 3 discusses the methodological framework of the study. It details model selection and architecture, dataset construction and pre-processing, training configuration, and model merging strategies.

Chapter 4 defines the experimental setup, including baseline systems, experimental conditions, and inference generation.

Chapter 5 details the framework and metrics used to quantify fluency, readability, and simplification quality.

Chapter 6 reports the experimental results, performance comparisons, merge strategy evaluations and ablation studies.

Chapter 7 discusses the findings and analysis of their ability to answer the stated RQs from section 1.2. It provides results interpretation, consideration for pedagogical implications, and addresses the study limitations and validity.

Finally, Chapter 8 concludes with summary of our main findings, outlines our contributions, and identifies directions for future work.

We also include a post-script addressing the potential and shortcomings of generative AI use in the pursuit of academic research.

CHAPTER 2

LITERATURE REVIEW

This literature review surveys prior research on TS with a focus on controllable simplification for educational use. It first reviews foundational approaches to simplification, from manual and rule-based methods to statistical and neural models. It then examines controllable simplification methods that target readability through explicit features, control signals, or prompting. Next, it reviews adapter-based fine-tuning and model-merging approaches as a modular alternative to retraining full models. Finally, it discusses the evaluation strategies used in simplification research, with particular attention to readability metrics and their limitations in L2 learning contexts.

2.1 Simplification of Natural Language Texts

To contextualize the present investigation, this section reviews prior work in text simplification, with a focus on sentence-level methods, model composition techniques, and evaluation strategies relevant to controllable simplification.

Manual TS. Systematic manual TS employs human experts referencing codified rule sets by which texts are analyzed at the sentence level and simplified in a consistent and structured manner. A taxonomy of controlled language (CL) rules has been proposed in O’Brien (2003), *Controlling Controlled English*, and contains the following four categories of rules:

1. Lexical rules influencing word selection
2. Syntactic rules simplifying intra-sentential complexity
3. Structural rules pertaining to the graphical layout and reducing information density
4. Pragmatic rules clarifying purpose or intent

Each of these categories is divided into its own sub-categories, depending on the CL rule set being employed, with as many as *60 sub-categories of rules* in the most complex rule sets. However, the author concluded from their survey that only one rule was common between each of the eight CL rule sets examined: that of promoting short sentences (O’Brien, 2003). This inconsistency highlights the drawbacks of subjective manual systems and serves as a potential argument in favor of statistical or neural automatic systems. At the same time, the reliance of manual TS systems for text simplification on human experts serves as a major bottleneck, in terms of both speed and cost. While often producing high-quality results, manual TS, being performed by native speakers and linguistic experts, is inherently limited by several factors: First, human judgments are prone to variability and subjectivity, leading to inconsistencies in the simplified outputs. Second, the process is time-consuming, as each text must be individually assessed and rewritten, making it impractical for large-scale applications. Third, the need for specialized expertise drives up operational costs, and the availability of qualified experts may be limited, especially for less widely spoken languages.

Rule-Based Automated TS. By automating the application of rule sets crafted for manual TS, computers have been able to partially alleviate the need for linguistic experts to perform TS by hand, though the output of these automated systems still requires verification. Although automated rule-based systems do not have the capacity to monitor and employ the pragmatic or aesthetic aspects of language that linguistic experts are able to account for, these rule-based approaches may still be structured as lexical simplification systems, syntactic simplification systems, or hybrid systems combining both methods (Al-Thanyyan and Azmi, 2021).

Statistical TS. Due to the increasingly rapid advancement of technology between the late 1900’s and early 2010’s, computer-based systems for automated statistical text simplification (STS) became increasingly practical and advanced. Though not without their own flaws, automatic TS systems follow in the path of statistical machine translation (SMT) techniques, being built around massive parallel corpora (i.e., aligned pairs of complex and simplified texts in the same language) and employing established reading metrics (see section 1.1.3 for more on readability scores) as heuristics (i.e., scoring functions used to estimate the quality or simplicity of a candidate output) for estimating text complexity (Al-Thanyyan and Azmi, 2021). In the same way that SMT utilizes parallel corpora to map words and phrases between languages, parallel corpora composed of complex-simplified sentence pairs compiled by human linguistic experts allow statistical models to learn mappings between words/phrases and corresponding simplified versions (Siddharthan, 2014). These learned mappings form the alignment template responsible for lexical phrase-based SMT (Och and Ney, 2004) and by extension, generalized syntactic tree simplification (Paetzold and Specia, 2013).

Neural TS. As in many fields of machine learning (ML) and natural language processing (NLP), NTS systems take the predictive power of statistical association as in statistical TS and enhances this power through the innumerable connections found in modern neural networks. NTS follows the many of the same principles as sequence-to-sequence (seq2seq) neural machine translation (NMT), where an attention-based encode-decoder neural architecture (e.g., recurrent neural network (RNN)) learns a sequence transformation mapping one language onto another (Bahdanau, 2014). The difference between STS and NMT is that, while NMT performs a sequence transformation from one natural language into another, NTS essentially performs monolingual translation, transforming complex text into simplified text in the same language.

TS with LLMs. In 2017 Vaswani et al. (2017) at Google dropped a bombshell on the NLP community in the form of their historic paper, Attention is All You Need. This marked the beginning of the nascent era of the large language model (LLM), revolutionizing myriad fields of research, TS being no exception. In contrast to the parallel corpora used to train increasingly large neural models, Feng et al. (2023) re-frame TS (which the authors refer to as sentence simplification, or SS) as a prompted input-output problem. Here, prompting is the use of an additional contextual instruction preceding the input string, and output is generated accordingly. Using GPT3.5 and ChatGPT, the authors were able to show that zero-shot learning (prompt contains instructions and input, but no examples) and one-shot learning (prompt contains instructions, input, and a single example simplification pair) outperformed state-of-the-art STS and NTS models. One alternative approach to n-shot learning is domain-specific fine-tuning, resulting in a so-called “expert language-model”. These expert models lack the generalizability of instruction-tuned models but make up for this lack of breadth with increased depth. Indeed, there is evidence that multi-task instruction-tuned models can suffer from task cross-contamination and lack the specialization power of expert models (Jang et al., 2023). These expert models can then be merged for improved performance on both familiar tasks, as well as generalizing to unseen datasets and tasks to a degree that instruction-tuned models are unable to surpass.

2.2 Controllable Text Simplification

Most TS systems focus on generating simpler outputs without explicitly controlling the level or nature of the simplification (Xu et al., 2015, Zhang and Lapata, 2017). But in pedagogical applications, we require grade or level alignment for the resources to be appropriate according to the students’ proficiency. For education, simplification cannot be a one-size-fits-all approach to complexity reduction; instead, we must employ a calibrated approach to approximate the needs of students at any given level (e.g., CEFR bands, school grades, etc.). This necessity motivates controllable TS research into models able to produce varied levels of complexity, given a control

sequence or other constraints. Viewed against this literature, the adapter-merging approach explored in this thesis should not be understood as a more precise alternative to explicit feature control. Rather, it is a modular alternative that trades some control precision for post-hoc compositional flexibility.

Control Tokens. One approach to controlling text generation for TS is the use of control tokens as markers added to the model input. Similar to how multilingual neural machine translation (MT) would use a target language token to condition generation during translation (Johnson et al., 2017), TS models may encode a discrete token to steer the readability level or other attributes. In 2020, Martin et al. proposed ACCESS, conditioning on linguistic features such as character length, dependency depth, and word frequency. Following Martin et al. (2020), Agrawal and Carpuat (2023) use predicted linguistic features corresponding to the desired grade-level output to condition the output of a fine-tuned T5 model. Both of these approaches utilize explicit control tokens during fine-tuning and inference to steer the model toward the desired output readability level. These explicit control tokens aid in interpretability and reproducibility, at the expense of requiring carefully curated parallel corpora tagged with the appropriate labels during fine-tuning. In circumstances where there is a paucity of data of this nature, fine-tuning approaches can struggle with generalizing across domains, languages, or readability levels.

Prompt Conditioning. In contrast to the explicit control tokens present during fine-tuning described above, advances in context window sizes have enabled the use of natural language prompt conditioning for controlling and influencing model outputs. A simple example of this is by prompting an instruction-tuned model to “rewrite the following sentence for grade 6”, allowing us to shift control from training to inference time (Zhou et al., 2023). As promising as this is, Sun et al. (2023) demonstrate that LLMs “struggle at fine-grained hard constraints, such as numerical planning and paraphrase generations,” when control is exercised only through abstract, high-level constraints. Building on this, Thillainathan and Koller (2024) propose a method for “fine-grained controllable text generation through in-context learning with feedback.” By explicitly specifying the linguistic features (e.g., dependency depth, dependency length, word count, and number of difficult words) and incorporating a feedback loop via external validator, their model improves accuracy over dependency depth and grade-level generation using only a small number of in-context examples. This approach demonstrates that explicit feature-level control can produce highly accurate grade-aligned rewrites, particularly when generation is guided by feedback and carefully specified linguistic targets. At the same time, these methods depend on external feature prediction, prompt design, and additional control mechanisms at inference time. The present thesis investigates an alternative tradeoff: whether post-hoc composition of grade-specific adapters can recover approximate readability control through parameter interpolation, even if that control proves less precise than explicit feature-based approaches.

2.3 Model Merging

As the field of NLP increasingly relies on large-scale pre-trained language models, the cost of adapting these models for specific downstream tasks has become a major bottleneck. Full fine-tuning of large foundational models like LLaMA3 and GPT-4 remains prohibitively expensive for most academic and educational institutions, requiring significant computational resources and storage for each new variant. Even modest domain adaptation via full model fine-tuning becomes impractical when the goal is to support a wide range of user-specific or task-specific behaviors, as is the case in personalized text simplification. This has led to the rise of Parameter-Efficient Fine-Tuning (PEFT) techniques such as LoRA, which allow practitioners to adapt LLMs to new tasks by training a small number of additional parameters while freezing the base model (Hu et al., 2021). LoRA drastically reduces the cost of this specialization while maintaining strong performance, enabling the training of multiple lightweight adapter modules for distinct tasks or domains. The key question is therefore not whether model merging is possible in general, but whether it yields a form of control that is useful for simplification, where small changes in output difficulty matter.

However, the emergence of numerous task-specific adapters introduces a new question: can these adapters even be recombined after training to approximate new behaviors, especially when data for the target setting is sparse or unavailable? In the context of graded text simplification, this question is especially relevant. Curating high-quality simplification data for every target grade level is resource-intensive, and intermediate levels (e.g., “grade 7.5” or “CEFR B2.1”) may not be represented at all. The present thesis tests whether weighted merging across adjacent grade-specific adapters can approximate such missing levels without requiring a separately trained adapter for every conceivable point on the readability scale. In fact, recent work has begun to explore this possibility. Techniques such as Branch-Train Mix (BTX) (Sukhbaatar et al., 2024) and AdapterFusion (Pfeiffer et al., 2021) propose merging multiple adapters trained on different tasks or domains to produce more generalizable or controllable models. Branch-Train Merge (BTM) trains separate branches for each task and merges them at inference time or during a secondary fine-tuning phase. AdapterFusion aggregates outputs from multiple adapters through a learned attention mechanism. These approaches show promise in improving performance and generalization on unseen tasks, but their utility for controlled generation (i.e., producing simplifications at a specific difficulty level) remains underexplored.

As it stands, most existing TS models operate under a binary or at best coarse definition of difficulty, aiming simply to make text “simpler” without modeling the continuous scale of readability complexity. Surveying the field confirms this trend: most systems are optimized for general simplification, with little to no capability to adjust to varying levels of difficulty beyond “original” and “simplified” (Garbacea et al., 2021). This coarse approach limits models’ ability to generalize to unseen or intermediate difficulty levels, especially in educational contexts where

such granularity is essential for supporting individual student needs. As shown by Crossley et al. (2007), simplified texts do not necessarily correspond to improved accessibility, and learners benefit more from texts that are appropriately aligned with their proficiency level than from undifferentiated simplifications. The end result is that while models fine-tuned to an arbitrary simplification objective may perform well on benchmark datasets, these models lack the means to target a specific learner proficiency or instructional level without complete retraining (Martin et al., 2020, Zhang and Lapata, 2017). Recent work has begun to explore LLMs for text simplification, including methods for both general simplification and control over target difficulty levels. Notably, Agrawal and Carpuat (2023) investigate prompt-based strategies to guide LLMs toward producing simplified outputs at defined levels of complexity, rather than treating simplification as a binary task. Similarly, Thillainathan and Koller (2024) extend this line of work by conditioning in-context generation on explicit linguistic features and grade-level annotations, enabling a form of target-level simplification that produces outputs aligned to specific readability bands. While these approaches differ in prompting strategies and supervision mechanisms, they both aim to move beyond coarse-grained “simple vs. original” dichotomies toward more nuanced, level-aware generation.

This thesis addresses this gap by testing whether graded adapter merging can provide a modular alternative to explicit feature control. Specifically, it evaluates whether LoRA-based adapters trained on reading-grade-labeled data can be merged through weighted averaging to approximate simplification levels not explicitly represented in training. The appeal of this approach lies in its efficiency: if successful, it would reduce the need to train a separate model for every target level. As the later chapters show, however, the resulting control remains coarse rather than precise.

2.3.1 Comparison of Merging Strategies

Several recent methods have been proposed for combining models trained on distinct tasks or domains, each with different implications for their modularity, generalization, and controllability. This section reviews four relevant merging paradigms in the context of controllable text simplification and motivates the adapter-based composition strategy used in this thesis.

Branch-Train-Merge (Li et al., 2022) involves training multiple task-specific adapters or model branches independently, followed by an additional optimization or gating step that fuses them into a unified model. This merging phase often improves generalization and downstream performance but comes at the cost of increased computational demands and reduced modularity. Additionally, the resulting merged model typically loses transparency with respect to the individual contributions of each component, which is an important limitation in educational contexts where interpretability and pedagogical alignment are crucial.

Branch-Train-Mix (Sukhbaatar et al., 2024), in contrast, avoids additional training by combining separately trained adapters at inference time via weighted interpolation. This approach retains full modularity and enables lightweight experimentation by combining separately trained adapters at inference time. In principle, it also offers a mechanism for interpolating between discrete control settings, which makes it attractive for graded simplification. Whether this adapter merger interpolation yields reliable or precise control over readability, however, remains to be seen.

AdapterSoup (Chronopoulou et al., 2023) selects a subset of pre-trained adapters based on their downstream performance, then merges them either greedily or through a learned mechanism. Although this approach can yield strong results for transfer or multitask generalization, it is not designed for interpolation. This distinction matters, because the present thesis asks whether adjacent readability levels can be approximated through composition, whereas AdapterSoup-style selection assumes that useful behavior can be recovered by choosing and averaging strong task specialists. The results in this thesis suggest that this kind of mix-and-match intuition is too optimistic for readability control: weighted combinations do not improve performance beyond the strongest fine-tuned adapters, and their interpolation effects remain relatively weak.

Model Merging by Uncertainty-Based Gradient Matching (Daheim et al., 2024) seeks to align task gradients during model combination using uncertainty measures. While theoretically grounded, this method is computationally intensive and requires access to task-specific data and training gradients; constraints that are incompatible with modular, parameter-efficient fine-tuning frameworks such as LoRA.

Post-hoc Adapter Merging. In contrast to the above approaches, this thesis adopts a modular training strategy in which a shared base adapter is first fine-tuned on all shared simplification data. From this baseline, separate grade-specific adapters are initialized and fine-tuned independently on their respective datasets. These adapters are then combined post-hoc using weighted averaging techniques, without additional fine-tuning. The motivation for this design is not to assume that parameter mixing will automatically yield precise readability control, but to test whether grade-specific simplification behavior can be recombined after training. Although structurally similar to BTX, this method differs in that adapter merging is performed only after all grade-specific models have been independently fine-tuned from a common initialization, rather than trained in tandem.

In summary, the purpose of the modular, two-stage adapter training and post-hoc merging strategy used here is to enable precise control over output difficulty with minimal retraining. It maintains transparency and compatibility with parameter-efficient methods, aligning well

with the pedagogical aims of L2 simplification. While inspired by BTX, the approach this thesis follows introduces key differences that support more interpretable and flexible control over readability during generation.

2.3.2 Dimensions of Merge Methods:

To make adapter merging more flexible and interpretable, this section introduces a set of merge methods organized by the primary design aspects they employ. These methods differ in how they combine model weights, and the goal here is to make those differences clear enough to support the decision to use a particular approach for TS.

Base Strategy refers to the fundamental approach used to merge models, which determines how their parameters or internal representations are combined. In a linear merge strategy, the parameters of multiple models are merged through a simple weighted average, as seen in Wortsman et al. (2022)’s Model Soup. The Task-Informed Embedding Similarity (TIES) method, proposed by Yadav et al. (2023), aligns and merges model layers based on similarity scores that are learned with respect to the target task. A concatenation (CAT) strategy merges models by stacking (concatenating) their parameters or intermediate representations, enabling hybridization across layers, as demonstrated by Ilharco et al. (2022). In magnitude-based merging (pruning), parameters are selected or weighted based on their absolute values, a technique adapted from pruning methodologies (Deep et al., 2024). Finally, Singular Value Decomposition (SVD) offers a compression-based merging strategy, reducing the dimensionality of combined models by factorizing large weight matrices into lower-rank approximations (Stoica et al., 2024).

Domain-Aware Representation Enhancement (DARE). Enriches the merge process by incorporating domain knowledge. When present, it modifies how task similarities or weight contributions are computed by selectively zeroing out less important parameters and rescaling those remaining (Yu et al., 2024).

SVD Augmentation. Whether SVD is applied after merging, typically to reduce redundancy or model size. Can be particularly useful when adapting to resource-constrained settings or when cross-domain robustness is the primary concern.

Pruning. Whether small weights are removed after the merge to make the model sparser.

Merge Method	Strategy	+DARE	+SVD
svd	svd	×	✓
linear	linear	×	×
cat	cat	×	×
ties	ties	×	×
ties_svd	ties	×	✓
dare_ties	ties	✓	×
dare_linear	linear	✓	×
dare_ties_svd	ties	✓	✓
dare_linear_svd	linear	✓	✓
magnitude_prune	pruning	×	×
magnitude_prune_svd	pruning	×	✓

TABLE 2.1: Merge methods arranged by strategy, $\pm$ domain adaptation, and $\pm$ SVD.

Table 2.1 (above) delineates HuggingFace’s available merge methods. Among the many merge methods available, we selected dare_ties as the primary experimental implementation in this thesis, as it combines two key features relevant to TS in the context of SLA:

Structural Alignment via TIES. By aligning layers based on similarity metrics, TIES preserves task-specific structural integrity. This is particularly beneficial for merging models targeting different readability levels, where syntactic and semantic patterns vary subtly but importantly.

Domain Sensitivity via DARE. Domain-Aware Representation Enhancement (DARE) enhances the alignment process by introducing domain-level contextual awareness, ensuring that merges account not only for model-internal features but also for domain-specific linguistic and topical attributes.

While the dare_ties_svd variant introduces an additional step of dimensionality reduction that can be beneficial in other contexts, it was not adopted here. SVD offers potential advantages in compressing models, removing noise, and improving inference efficiency. However, these benefits come at the cost of interpretability and control. For text simplification, particularly in SLA, preserving fine-grained distinctions in linguistic structure is crucial. SVD-based compression risks removing low-variance yet pedagogically relevant patterns, undermining the goal of fine-grained readability interpolation. Moreover, the increased abstraction of SVD-transformed models harm explainability, a primary concern in educational applications. For these reasons, dare_ties is favored for its transparency, granularity, and pedagogical applicability.

In summary, we selected Domain-Aware Representation Enhancement with Task-Informed Embedding Similarity (DARE.TIES) as the primary merging strategy because it offers the most plausible balance between interpretability and flexible readability control within our experimental framework. More importantly, preliminary experiments show that alternative merge operators

consistently produce catastrophic failures, while DARE_TIES is the only method that remains stable enough to support systematic experimentation. The choice is therefore motivated not only by theoretical considerations, but by empirical necessity.

2.4 Simplification and Readability Metrics

Evaluating TS systems requires balancing partially competing objectives: while the primary goal is to reduce text complexity and improve readability, proper simplifications should maximize meaning preservation while aligning with the intended linguistic complexity. Research into evaluating TS relies on a combination of surface-level metrics, as well as reference-based and psycholinguistically motivated measures. Although no single metric can provide a complete picture of simplification adequacy, this section reviews standard methods and examines their limitations.

2.4.1 Surface Metrics

Surface readability metrics estimate text difficulty using shallow lexical and syntactic measures. The FKGL formula (Kincaid (1975), formally defined in Chapter 5.2) predicts grade-level readability using a combination of average words per sentence and average syllables per word. Because of its interpretability and computational simplicity, FKGL is commonly used in TS research during model training and evaluation.

Despite its widespread use, FKGL has well-documented limitations when applied to TS and language learning contexts. Davison and Kantor (1982) argue that surface-level formulae like FKGL often fail to capture the actual cognitive demands placed on readers, particularly when sentence length reductions mask underlying syntactic complexity or loss of cohesion. Simplification operations such as sentence splitting may reduce FKGL scores while introducing more complex inter-sentential discourse relations that challenge L2 readers. Beck et al. (1984) further note that certain simplification strategies aimed at improving text coherence (e.g., adding referential phrases, clarifications, or explicit discourse markers) can paradoxically lead to higher FKGL scores, since these additions increase sentence length or syllable counts. As a result, improvements in comprehensibility may not be reflected in lower readability scores, highlighting a key limitation of relying solely on FKGL for evaluating simplification effectiveness.

Additionally, Tanprasert and Kauchak (2021) demonstrate that optimizing for FKGL can lead to improper simplification strategies by introducing uncommon lexical items or splitting sentences with no regard for meaning or comprehensibility preservation. These findings emphasize that FKGL approximates structural simplification, but does not reflect semantic adequacy or

discourse clarity. As such, FKGL should be interpreted as a coarse-grained indicator of surface-level simplification, rather than a comprehensive measure of comprehensibility or pedagogical suitability. When taken together, these limitations further motivate the use of complementary evaluation metrics, as described in the following sections.

In addition to FKGL, several alternative readability metrics exist, including the Gunning Fog Index (Gunning, 1952), the SMOG Index (Mc Laughlin, 1969), the Dale-Chall Formula (Dale and Chall, 1948), and the Fry Readability Formula (Fry, 1968). While these formulae capture different aspects of surface-level complexity, they remain highly correlated and similarly limited in their ability to reflect deeper syntactic, lexical, and semantic features relevant for SLA (Beck et al., 1984, Davison and Kantor, 1982). More recent approaches incorporating psycholinguistic variables or syntactic features exist (Crossley et al., 2008, Siddharthan, 2014), but often require additional linguistic resources or annotations not available for all datasets, as well as significant pre- and post-processing, limiting the scalability and general applicability of these methods in real-world TS systems.

2.4.2 Reference-based Metrics

Since TS can be viewed as a monolingual variant of MT, many standard MT evaluation metrics have been applied to this task, including BLEU (Papineni et al., 2002), METEOR (Banerjee and Lavie, 2005), NIST (Przybocki et al., 2010), LEPOR (Han et al., 2012), and Word Error Rate (Klakow and Peters, 2002). These metrics generally measure n-gram overlap or edit distance between the system output and one or more reference simplifications. While widely used in MT, they correlate poorly with human judgments of simplification quality, as they do not explicitly account for the distinct operations involved in simplification, such as necessary deletions, insertions, or rephrasings (Alva-Manchego et al., 2021, Xu et al., 2016).

More recent embedding-based metrics such as BERTScore F1 allow for semantic similarity comparisons between system output and reference texts (Zhang et al., 2020). However, these approaches primarily capture meaning preservation and are less informative about whether appropriate simplification has occurred. Given these limitations, this study supplements BERTScore F1 with SARI from Xu et al. (2016) as the primary evaluation metric for simplification quality. SARI is explicitly designed for text simplification and separately evaluates the quality of words that are added, kept, or deleted relative to both the input and the reference. This aligns closely with the nature of the simplification task, which involves selectively removing, rephrasing, or preserving information while reducing complexity. SARI has also been shown to correlate more strongly with human judgments of simplification quality than traditional MT metrics, making it particularly well-suited for evaluating systems where simplification adequacy is not necessarily reflected by n-gram overlap alone.

2.4.3 Psycholinguistic Metrics

Moving away from surface-level considerations, psycholinguistically motivated readability metrics attempt to more directly approximate the cognitive processing involved in text comprehension. Not only do word frequency and familiarity directly influence comprehension (Nouri and Zerhouni, 2018), but more opaque structural features such as dependency depth and dependency length also correlate with working memory load during sentence parsing (Gibson et al., 2000).

More recent approaches to controllable text generation incorporate these features as generation parameters. Thillainathan and Koller (2024) demonstrate that, using few-shot learning, LLMs can perform accurate rewrites to match specified values for dependency depth, dependency length, and lexical difficulty. While this research targets text comprehension more directly than simple syllable counts and other surface-level proxies for text difficulty and readability, these measures lack a standardized (grade-level) scale for comparison against educational frameworks such as CEFR or the U.S. school grade system.

2.4.4 Correlation with Comprehension

The “elephant in the room” is the degree to which automatic readability metrics do or do not correlate with human comprehension. Surface-level readability formulae such as the FKGL were originally developed and validated against standardized comprehension tests for a specific test population (Kincaid, 1975). However, recent work shows that these results do not generalize consistently to other learners or text genres (Tanprasert and Kauchak, 2021). Moreover, a model’s improvement as measured by these automatic metrics says nothing about improvements in human understanding or adequacy ratings (Xu et al., 2016). For L2 learners especially, prior exposure and lexical familiarity play important roles in determining understanding, which are simply not captured by these methods (Krashen, 1982). While human evaluation remains the gold standard, automatic metrics persist and excel due to their scalability and reproducibility. With this in mind, we can partially mitigate the shortcomings of any single metric when used in isolation by combining these metrics in our analysis and evaluation of model behavior.

2.5 Text Simplification for L2 Pedagogy

Within L2 pedagogy, TS maintains a unique role: when compared to more general simplification tasks, which prioritize readability reductions in a somewhat more general sense, TS for SLA must balance input quality, comprehensibility, and ideally developmental progression. In supporting the unique needs of L2 learners, simplification must support language acquisition through controlled adaptation, not simply structural complexity reduction. Thus the novelty of this

work lies not in proposing a new merge operator, but in testing whether adapter merger can approximate pedagogically meaningful difficulty levels in a simplification setting where precise supervision is expensive and intermediate target levels are often unavailable.

2.5.1 Comprehensible Input and Developmental Appropriateness

As stated in section 1.1.1, SLA theory places a heavy emphasis on input that is comprehensible while still being slightly beyond the learner’s current competence. The $i+1$ notion from Krashen’s Input Hypothesis tells us that input must be understandable while still containing novel forms. This hypothesis has its supporters as well as detractors, but the notion that target language input must align with learner proficiency is not debated in L2 pedagogy. While graded readers and course-specific instruction materials organize their content according to standard features such as vocabulary size, syntactic complexity, and discourse structure, these materials rely on manual adaptation or creation of level-appropriate texts according to course proficiency levels. Automatic TS opens the door to the possibility of dynamically adapting texts to better align with an individual learner’s needs, as long as these systems can control for the relevant linguistic features.

2.5.2 Simplification Operations and Pedagogical Function

Research in TS focuses on three primary modes of transformation: lexical simplification, syntactic simplification, and discourse-level modification (Alva-Manchego et al., 2021). This section examines these modes in relation to their pedagogical applications and considerations.

Lexical Simplification. Lexical Simplification involves replacing low-frequency, difficult, or niche vocabulary with shorter, more frequently appearing alternatives. In particular, lexical familiarity strongly influences L2 learner comprehension and processing speed (Gibson et al., 2000). By replacing infrequently used terms with more common vocabulary, simplified texts for SLA reinforce vocabulary acquisition through repeated exposure to a more limited set of lexical entries. We can approximate lexical difficulty using word frequency thresholds or specially curated word lists such as the Dale-Chall list (Dale and Chall, 1948). By cutting off the long tail of the Zipfian distribution (Zipf, 1935), we dramatically increase readability while preserving meaning, at the expense of some degree of nuance and artistic intent in the original text.

Syntactic Simplification. Syntactic Simplification involves structural changes to the text to lower reading difficulty. This typically includes sentence splitting, clause reduction, and the reordering of complex clauses. Psycholinguistic research links dependency length and depth

to processing difficulty (Gibson et al., 2000), and these kinds of deeply nested structures and long-distance dependencies can hinder comprehension for L2 learners. While controllable text generation frameworks are moving in the direction of explicit feature modeling for linguistic features such as dependency depth and length (Martin et al., 2020, Thillainathan and Koller, 2024), we can approximate syntactic complexity without explicit modeling using reading scores that include measures of sentence length, such as FKGL.

Discourse and Cohesion. Discourse and Cohesion considerations go beyond lexical and syntactic simplification, by ensuring content remains coherent after simplification. Sentence splitting alone does not necessarily reduce the processing burden of a text, especially when deeper structural relations remain unresolved. For this reason, coherence-preserving operations such as pronoun resolution and the addition of discourse devices (e.g., logical connectors) are also important for ensuring that simplified texts remain understandable in context. Although similarity metrics such as SARI and BERTScore F1 provide evidence for preservation of meaning and coherence, as always, human evaluation remains the gold standard for assessing simplification adequacy.

2.5.3 Linking Pedagogical Objectives to Evaluation Criteria

As stated above, special consideration must be given to TS designed for SLA, aligning the specific pedagogical objectives to our evaluation criteria. With this in mind, lexical accessibility and syntactic processing motivate the use of vocabulary readability formulae like FKGL. The requirement for meaning preservation motivates the use of reference- and embedding-based methods for similarity measurement, i.e., SARI and BERTScore F1, respectively.

CHAPTER 3

METHODOLOGY

This chapter describes the methodological framework used to train and evaluate the controllable TS models investigated in this thesis. The goal of the methodology is to determine whether readability-specific adapters can be combined through parameter merging to approximate intermediate difficulty levels without training separate models for every grade.

The chapter proceeds as follows. First, we describe the dataset construction process, including readability annotation and filtering procedures used to improve simplification quality. Next, we present the model architecture and PEFT configuration based on LoRA. We then outline the training strategy used to produce grade-specific adapters, followed by the adapter merging procedures used to test readability control through (weighted) model composition. Finally, we describe checkpoint selection criteria and implementation details necessary for reproducibility.

3.1 Experimental Pipeline

The methodology follows a staged experimental pipeline designed to separate three questions: (i) how well grade-specific adapters perform when trained individually, (ii) whether a shared initialization improves the compatibility of those adapters for later merging, and (iii) whether merged adapters can approximate target readability levels more effectively than individually fine-tuned adapters.

The pipeline begins with dataset construction. After additional cleaning and filtering, the simplification corpus is annotated with FKGL-derived grade labels and partitioned into grade-specific subsets spanning Grades 2 through 12. These graded subsets provide the basis for all later adapter training and evaluation.

From this point, the experiments branch into two training conditions. In the shared warm-up condition, a single LoRA adapter is first trained on the full multi-grade simplification dataset.

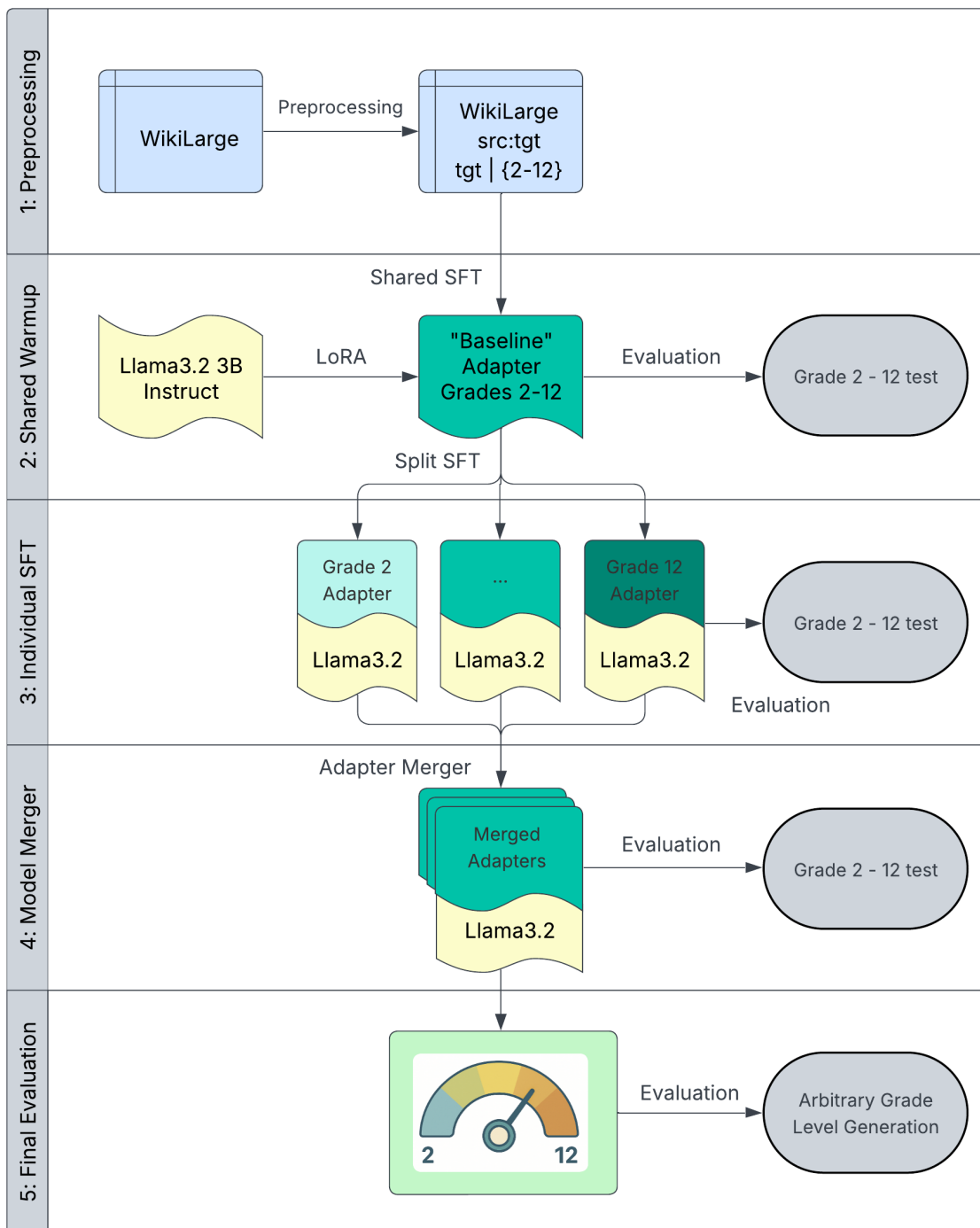
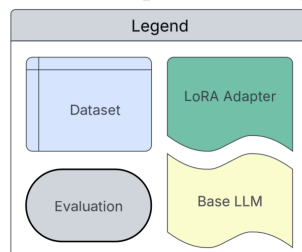


FIGURE 3.1: Experimental Pipeline



This shared adapter serves as a common initialization point for all subsequent grade-specific adapters. Eleven grade-specific adapters are then derived from this checkpoint and fine-tuned independently on their respective grade-level datasets. The motivation for this branch is that a common initialization may keep the resulting adapters more structurally aligned, which in turn may improve the stability during parameter merging. In the off-the-shelf (OTS) condition, the grade-specific adapters are instead trained directly from the frozen base model without the shared warm-up stage. This branch serves as a comparison point for determining whether shared initialization actually improves downstream readability control or merge compatibility. As in the shared warm-up branch, eleven grade-specific adapters are trained independently, one for each target grade.

These two branches produce two parallel adapter families: **graded-from-shared-warmup** and **graded-from-off-the-shelf**. Each family is evaluated first in isolation as a set of individually trained grade-specific adapters. This allows the thesis to establish how accurately each training strategy captures target readability before any merging is introduced. After individual evaluation, both adapter families are used in parameter-merging experiments. For the shared warm-up branch, the grade-specific adapters are merged under a range of weighting schemes, merge window sizes, and adapter-selection configurations. These experiments form the main body of the thesis, as they test whether readability control can be improved or interpolated through the weighted composition of grade-specific adapters. For the OTS branch, a smaller set of merge experiments is conducted using the same core weighting strategies, primarily to determine whether the shared warm-up stage improves merge behavior relative to adapters trained directly from the base model.

This structure allows the experimental design to isolate the contribution of each stage. The individual-adapter evaluations address whether discrete grade-specific training already provides useful readability control. The comparison between shared warm-up and OTS training addresses whether common initialization improves compatibility across adapters. The merge experiments then test whether post-hoc composition can recover approximate intermediate readability levels and/or improve alignment with target grades.

Figure 3.1 summarizes the overall experimental pipeline used in this study.

3.2 Dataset Construction

The training data for simplification in this work begins life as a preprocessed version of the Zhang and Lapata (2017) WikiLarge dataset graciously provided by Thillainathan and Koller (2024). The original WikiLarge corpus is a large-scale collection of sentence-aligned complex and simplified English sentences, selected for its domain diversity and scale. The Thillainathan

and Koller (2024) version includes cleaned and de-tokenized sentence pairs annotated with estimated grade levels and linguistic feature values, enabling more structured experimentation with readability control. However, prior research has raised concerns about the alignment quality of simplifications in Wikipedia-based datasets, including WikiLarge. Citing issues such as semantic drift, minimal simplification, or missing output altogether, Xu et al. (2015) shows that as much as 50% of the corpus does not adequately satisfy the task of sentence simplification. Preliminary inspection of the preprocessed dataset confirms that many such issues persist, motivating further cleaning and filtering to improve training quality.

Grade-Level Annotation. We derive grade-level labels using the FKGL readability metric, as defined in Chapter 5.2. The FKGL formula produces a continuous score that ranges from 0 to positive infinity. For each simplified sentence, we compute its FKGL score and round the value to the nearest integer. Because FKGL can produce values above the traditional K-12 scale, we cap all scores greater than 12 at “grade” 13 prior to filtering. We then exclude instances labeled as Grades 0 and 1 due to small sample sizes and poor data quality, and we remove Grade 13 instances, as they fall outside the intended K-12 grading framework of this study. The resulting dataset spans Grades 2 through 12, which constitute the dataset groupings and grade targets used during training and evaluation.

3.2.1 Data Cleaning Procedure

To improve the quality of the training data, we apply an automated filtering procedure to remove sentence pairs that do not represent valid simplifications. Prior work has shown that the WikiLarge dataset frequently contains misaligned pairs, missing simplifications, or minimal edits that do not meaningfully reduce complexity (Xu et al., 2015). Preliminary inspection confirmed that such issues remain present in the preprocessed dataset used in this study.

We therefore construct a binary classifier to identify and remove low-quality simplification pairs. Each source-target pair is represented using three features designed to capture semantic correspondence and simplification behavior.

1. **Semantic Similarity:** By encoding the source and target sentences with spaCy’s (Honnibal and Montani, 2015) `tok2vec en_core_web_sm` embeddings, we obtain a cosine semantic similarity score between the source and target sentences between $(-1, 1)$. Semantic similarity indicates whether the source and target sentences contain approximately the same information.
2. **Length Ratio:** Dividing the word count of the target simplification by the word count of the source sentence, results in a score between $(0, \text{inf})$. A score < 1 indicates that the

target simplification is shorter than the source sentence. The original dataset contains many empty or one-word target simplifications; thus length ratio identifies sentences that are inadequately reduced or missing. A score > 1 indicates the target is longer than the source, which can happen when complex sentences and concepts are broken up into multiple sentences or concepts that are individually simpler.

3. **Grade Difference:** The goal of the simplification dataset is to take an original text and reduce its complexity. By subtracting the FKGL grade of the source sentence from the target sentence, we identify whether the target sentence is a “simplification” of the original source sentence according to our primary evaluation metric. A negative grade difference indicates that the text has become easier to read, while a positive score indicates that a text has become more difficult, according to our grade scale.

To estimate the reliability of these features for detecting problematic examples, we manually annotated a random sample of 1,000 sentence pairs drawn from the dataset. Each pair was labeled as either a valid simplification (1) or an invalid or unrelated transformation (0). A pair was labeled as valid when the simplified sentence preserved the meaning of the source sentence while reducing linguistic complexity. Pairs were labeled as invalid when the target sentence was missing or unrelated.

The labeled dataset was used to train a Bernoulli Naïve Bayes (BNB) classifier using a 90:10 train-test split. On the held-out test set, the classifier achieved an F1 score of 0.95, indicating that the three features capture sufficient signal to distinguish valid simplifications from noisy or misaligned pairs.

After validating the classifier, we applied it to the full dataset to automatically remove examples predicted to be invalid. Instances classified as label 0 were discarded prior to constructing the final grade-specific datasets used for training.

Filtering Outcome. Before filtering, the dataset contained approximately 166,000 source-target sentence pairs. After applying the classifier-based filtering step, 124,000 pairs remained, corresponding to approximately 75% of the original data. The remaining examples were then partitioned into grade-specific subsets for subsequent training and evaluation.

Additional details on feature computation, labeling protocol, classifier training, and filtering thresholds are provided in Appendix C.

3.3 Model Architecture

Backbone Model. We conducted all experiments using the LLaMA3.2-3B-Instruct model from Meta AI (Meta AI, 2024), accessed through the Hugging Face (HF) Transformers library (Wolf et al., 2020), and fine-tuned using the open-source LLaMA-Factory v2 training framework (Zheng et al., 2024). Small by today’s standards, we chose the 3B version of LLaMA3 for its balance between performance and resource efficiency, allowing for rapid iteration and experimentation across multiple training configurations.

LoRA Configuration. In this section, we describe the architectural configuration of the simplification model used in our experiments. We fine-tune the LLaMA3 base model using LoRA. We specify the backbone architecture, LoRA hyperparameters, injection strategy, and resulting parameter counts. We load the model in `bf16` precision and freeze all backbone parameters during fine-tuning.

The model configuration is summarized below:

Parameter	Value
Hidden size (d)	3072
Intermediate size	8192
Number of layers (L)	28
Number of attention heads	24
Number of key-value heads	8
Head dimension	128
Vocabulary size	128,256
Maximum position embeddings	131,072

TABLE 3.1: Configuration of the LLaMA-3.2-3B-Instruct backbone model.

LoRA re-parameterizes a weight matrix $W \in \mathbb{R}^{d \times k}$ as:

$$W' = W + \Delta W, \quad \Delta W = BA, \quad \text{where} \quad A \in \mathbb{R}^{r \times k}, \quad B \in \mathbb{R}^{d \times r}.$$

By freezing W and optimizing only A and B , the effective update is scaled by $\alpha/r = 16/8 = 2$, which controls the magnitude of the injected adaptation.

We apply LoRA with the following hyperparameters:

Hyperparameter	Value
Rank (r)	8
Scaling factor (α)	16
Target modules	All linear projection layers
LoRA dropout	0.05

TABLE 3.2: LoRA configuration used for fine-tuning LLaMA-3.2-3B-Instruct.

Applying LoRA to all linear projection layers in the 28-layer LLaMA3 model introduces approximately 10.55 million trainable parameters. Given that the full model contains approximately 3 billion parameters, LoRA updates only about 0.35% of the total model parameters during fine-tuning, while keeping all remaining weights frozen. This parameter efficiency greatly reduces memory requirements and computational costs, while preserving the pretrained model’s linguistic competence.

3.4 Adapter Training Strategy

The training pipeline follows a two-stage adapter training strategy. First, a shared simplification adapter is trained using the combined dataset spanning Grades 2-12. This stage establishes a common initialization for subsequent grade-specific adapters.

In the second stage, separate adapters are initialized from the shared checkpoint and fine-tuned on grade-specific datasets. Each adapter, therefore, specializes in producing simplifications aligned with a particular readability level while preserving parameter compatibility with the other adapters. This shared initialization is important because it reduces parameter divergence between adapters, which improves the stability of later merging efforts.

The objective of this merging phase is to evaluate whether weighted adapter combinations can support precise control over output readability, generalize to unseen data, and interpolate between difficulty levels. This approach enables flexible experimentation with weight configurations while preserving modularity and transparency. By decoupling training from merging, this method supports interpretability and efficient adaptation to new readability targets, which are objectives otherwise not well served by prior fusion- or gating-based merge techniques.

3.4.1 Training and Merging Pipeline

Step 1: Dataset Cleaning and (Re)Splitting. The process begins with additional data cleaning and grade-level re-splitting. Following the use of a BNB classifier, as detailed in the previous section, the combined filtered dataset will again be subdivided by grade level. Each graded subset is split into training, development, and test sets. For consistency across evaluations, we anchor the development and test set sizes to the smallest grade-level subset using an 80:10:10 ratio, and adjust to the nearest multiple of 32 to enable consistent batching. For example, if an 80:10:10 ratio of the smallest graded subset results in train:dev:test split sizes of 4096:512:512, then 512 is the size of the development and test sets for each grade, with the remaining samples comprising their respective training sets.

Step 2: Baseline Adapter Training. We train a single baseline adapter on the combined training data from all grades (2-12), using LLaMA3 as the frozen base model. We construct the baseline development and test sets by sampling proportionately from each grade’s respective development and test splits.

Step 3: Grade-Specific Adapter Training:. We initialize eleven grade-specific adapters (grades 2-12) from the above shared baseline and fine-tune the grade-specific adapters on their individual grade-level datasets. This branching approach keeps the adapter weights structurally aligned to minimize interference when merging.

Step 4: Merge Method Selection. We first compare several parameter merging operators to determine which method provides the most stable readability behavior. Each merge combines the eleven grade-specific adapters using equal weights as a baseline. We evaluate the various candidate merge methods (see section 2.3.2 for comparisons), focusing on linear parameter averaging and DARE_TIES alignment. Based on these comparisons, the DARE_TIES merge operator provides the most stable readability behavior while avoiding the catastrophic parameter interference observed with naive linear averaging. For this reason, DARE_TIES is used as the primary merge operator in subsequent experiments.

Step 5: Manual Weighted Adapter Experiments. After selecting the best merge strategy, we test several manual weighting schemes to assess which different weighting configurations best steer the output readability.

These approaches include:

1. Uniform weighting across all adapters
2. Proximity-based weighting, emphasizing adapters closest to the target grade
3. Target exclusion experiments where the target-grade adapter is omitted
4. Weight scaling experiments (halved, doubled, and tripled weights) used to examine merge stability under different magnitudes
5. Random weighting baselines with different average magnitudes

3.5 Checkpoint Selection and Stability

We employ a two-stage fine-tuning procedure consisting of (1) a shared warm-up phase across all grades and (2) grade-specific adapter fine-tuning. Checkpointing and model selection follow different protocols in each stage to ensure stability and prevent premature selection.

3.5.1 Stage 1: Shared Warm-up

In the first stage, we fine-tune a shared LoRA adapter using the combined dataset spanning Grades 2-12. This warm-up stage runs for 5 epochs.

We checkpoint the model every 1,768 training steps, resulting in 10 checkpoints per epoch. This allows us to monitor training dynamics during the shared warm-up phase. The choice to checkpoint only 10 times per epoch is driven by the need for full generation on the development set during evaluation. This causes evaluation times to greatly outpace training times, in spite of the fractional size of the evaluation datasets. Across training runs, validation performance typically improves rapidly during the first epoch before plateauing. For this reason, shared warm-up checkpoint selection based on validation readability alignment provides a more reliable selection criterion than fixed-epoch training.

3.5.2 Stage 2: Grade-Specific Fine-Tuning

After completing the shared warm-up phase, we initialize separate LoRA adapters for each grade using the weights from the warm-up stage. We then fine-tune each adapter exclusively on target grade-matched training data for 10 epochs.

During this stage, we checkpoint once per epoch. This design ensures that no checkpoint is considered before completing a full pass over the grade-specific dataset, and that graded datasets of different sizes are evaluated the same number of times. By restricting checkpointing to epoch boundaries, we prevent grade adapters from selecting early partial-epoch checkpoints that may reflect unstable data subsets, at the cost of having different effective total samples using the same number of training epochs, depending on dataset size. Performance at different stages will be discussed in Chapter 6.

3.5.3 Checkpoint Selection Criterion

We compute the following validation metrics:

- Primary metric: change in Flesch-Kincaid Grade Level (ΔFKGL)
- Secondary metrics: BERTScore F1, SARI, perplexity

Checkpoint selection is based solely on the primary validation metric. Secondary metrics serve as guardrails against improper simplification behaviors, as well as more general indications of fitness and adequacy.

Best Checkpoint Selection Criterion. For each model, we select the checkpoint that minimizes the difference between FKGL scores of the generated text and reference simplifications on the corresponding development set.

Training Dynamics. Across grades, we observe that validation performance typically improves rapidly during the first epoch before plateauing. We do not observe catastrophic divergence or unstable loss spikes under the selected hyperparameter configuration.

Runtime. Training our grade-conditioned models requires approximately:

Training Phase	Time
Shared warm-up base model	9.00 hours
Average training time per model	0.75 hours
Total training time (all grades, incl. warm-up)	17.25 hours

TABLE 3.3: Training runtime statistics

Generation Settings. During evaluation, simplifications are generated using greedy decoding without sampling. This deterministic decoding strategy ensures that differences in output readability arise from model behavior rather than stochastic decoding variability.

3.6 Reproducibility

This section contains documentation of all the implementation details necessary to reproduce the experiments in this study, including a git repository containing source code, configuration files, pre-processing scripts, and evaluation scripts.

Code and Versioning. We implemented all experiments in Python using the HuggingFace Transformers ecosystem. All experiments reported in this thesis correspond to the repository state identified by the commit hash listed. We release the full code-base at <https://github.com/WilliamPLaCroix/LLaMA-Factory>, commit hash: `c14d823`. The repository contains: Data pre-processing scripts, LoRA fine-tuning configuration files, training and evaluation pipelines including automatic metric scripts for FKGL, SARI, BERTScore F1, perplexity, and Hyperparameter specifications.

Software and Library Versions. We conducted experiments using Python 3.10.16, PyTorch 2.5.1, Transformers 4.50.0, and PEFT 0.15.0. Training relied on CUDA 12.8 with cuDNN 9.1.0.70. Dataset processing used the HuggingFace Datasets library (3.2.0) and Evaluate (0.4.3).

Reproducibility. All pre-processing steps, dataset filtering decisions, and grade-level annotation procedures are deterministic, given the provided scripts and seeds. The released repository contains sufficient information to replicate dataset construction, model training, and evaluation results reported in this thesis.

We conduct training on a single NVIDIA H100, loading the model with `bfloat16` precision and training using LoRA-based fine-tuning. We set all random seeds globally to 42 to ensure deterministic behavior across runs. We also disable nondeterministic CUDA operations where applicable. No model parallelism beyond standard device placement is used.

CHAPTER 4

EXPERIMENTAL SETUP

4.1 Research Objectives

Before detailing the experimental setup, it will be helpful to frame our goals and experiments by restating the primary objectives of this thesis. This thesis investigates whether control over grade-specific simplification output is achievable by means of the weighted merger of individually pretrained adapters, without resorting to retraining or further fine-tuning. The work addresses the issue of controlling text output complexity at arbitrary readability levels along the FKGL scale. The experimental design, therefore, focuses on three core objectives, as detailed in Section 1.2:

- **Research Question 1 (RQ1):** Do sentence simplification adapters trained on discrete grade levels reliably generate output that corresponds to their target grade level, as measured by standard readability metrics?
- **Research Question 2 (RQ2):** Can combining grade-specific adapters through weighted parameter merging improve the system’s ability to generate text at intended readability levels?
- **Research Question 3 (RQ3):** Can a learnable weighting scheme over grade-specific adapters enable the generation of text at arbitrary intermediate readability levels not explicitly represented in training data?

Each of these questions requires its own experimental validation.

First, we need to establish whether graded adapters are capable of capturing the structural differences between outputs at different readability levels (i.e., syllables per word and words per sentence). We do this by testing the individual graded adapters against a common baseline, detailed below. Without establishing this capability, interpolation and model merger remain unmotivated.

Second, we determine whether merging and combining various graded adapters produces predictable changes in the complexity of the output text. We must determine whether weighted merger of these adapters produces superior results to naive simple merger where adapters are merged using equal weights. We must also determine what the optimal weighting scheme is for combining adapters along three axes: we must establish the dominant merge method from among those shown in table 2.1, both in terms of which adapters are selected, as well as the scaling magnitude of the included adapters’ weights. Our experimental setup therefore must systematically assess mixture combinations and weights, which we refer to as “weighting schemes.”

Third, inter-grade interpolation poses a more challenging claim, both in terms of implementation feasibility, as well as assessment reliability. This was beyond the scope of the current research, and will be discussed in more detail in Chapter 8.

Across all three experimental stages, assessment focuses on optimizing ΔFKGL , while SARI, BERTScore F1, and perplexity are recorded to guard against improper simplification behaviors while ensuring fluency and meaning preservation remain high. The following sections detail the specific procedure and setup involved in each of these experimental stages.

4.2 Baseline Systems

We first test an OTS LLaMA3 base model that has not undergone additional fine-tuning to establish zero-shot performance as our experimental baseline. For each dataset of target grade n , we generate simplifications for every sentence in the dataset individually, using the following prompt, followed by the source sentence, formatted in accordance with the LLaMA3 chat template (examples of the full prompt template including sources, targets, input tokens, and predictions provided in Appendix B):

Rewrite this Input sentence to make it easily understandable by students in Grade n

4.3 Shared Warm-up Model

In the second stage, we introduce a shared warm-up model. To obtain this model, we fine-tune the OTS base model on the full graded training corpus without first separating data by target grade. This provides a shared initialization point for all subsequent grade-specific adapters. This shared training stage is intended to reduce divergence between adapters and mitigate the potential for catastrophic forgetting during adapter merging in the following experimental phases.

The shared warm-up model is evaluated using the same prompting scheme as the OTS model. Its performance serves as a task-adapted baseline when compared against the OTS instruction fine-tuned base model, as well as isolating the effects of general simplification training from the effects of further grade specialization.

4.4 Individually Trained Graded Adapters

We next train separate LoRA adapters for each discrete target grade level. Each graded adapter is fine-tuned on all source grade examples corresponding to its assigned target simplification grade: *source grade* $m \rightarrow$ *target grade* n , $| m \in \{1, \dots, 13\}, n \in \{2, \dots, 12\}$. These adapters are evaluated individually to answer RQ1: whether discretely trained grade-level adapters produce outputs that reliably match their intended readability level.

To assess the efficacy of the shared warm-up initialization, we train two variants of all graded adapters for target grades 2 through 12:

- **Graded from shared warm-up:** adapters trained following initialization from the shared warm-up model
- **Graded from OTS:** adapters trained directly from the original base model without the shared warm-up stage

4.5 Merged Adapters

After evaluating graded adapters when tested individually against their respective target grades, we investigate several adapter merging methods before proceeding with the most promising merge method. We test each merge method by merging all graded adapters and testing each target grade separately to determine which merging method(s) best preserve readability control and simplification quality.

4.5.1 Adapter Merging Methods

We evaluate all adapter merging strategies provided by the HuggingFace merging framework via `add_weighted_adapter()`, summarized in Table 2.1. In addition to their primary merging methodology (e.g., linear, concatenation, TIES, etc.), these methods also differ along two principal dimensions: (i) whether they apply DARE, and (ii) whether they incorporate an SVD-based compression step. Again, we ignore SVD compression setups in this research.

Accordingly, we evaluate the following merge operators:

```
linear
ties
dare_linear
dare_ties
magnitude_prune
```

We begin by applying each method to all 11 graded adapters and test against each target grade level, as above with the baseline systems. Preliminary results bore out our prediction that DARE-TIES would be the dominant strategy, and indeed all other merge methods resulted in catastrophic failure (e.g., $20 < \Delta\text{FKGL} < 200$) in every tested configuration (see Appendix B for full examples of failure cases). Catastrophic failure in this context refers to consistent collapse into one or more of the following behaviors: non-language character output, copying and repetition of the source sentence without modification, complete breakdown of intelligibility, uncontrolled divergence from the requested grade level. These effects appeared consistently across linear, TIES, linear DARE, and magnitude-pruning variants. The results suggest that most merging operators introduce competitive and destructive parameter interference when applied to adapters in this context. For the remainder of the experimental setup, we retain a small number of `linear` merge experiments as a baseline merge method for comparison but pursue the remainder of our experimentation using DARE-TIES as the lone merge method worth exploring.

4.5.2 Merge Permutations

After identifying DARE-TIES as the strongest merging method, we conduct systematic permutation experiments to explore the interaction between:

- Weight aggregation and scaling methods
- Number of adapters included in the merge
- Adapter grade vs. target grade adjustments

With various weighting strategies, including:

- **Uniform:** equal contribution from each selected adapter
- **Proximity:** weights inversely proportional to their distance from the target grade
- **Random:** randomly sampled weights for robustness comparison

Our experimental configurations follow a structured naming scheme (see Appendix A for the full list of experimental configurations):

`[method]_ws@[window-size]_w@[weighting]_[aggregation]`

Window Size. The parameter window-size (`ws@`) indicates number of adapters merged, where the window is the number of adapters on either “side” of the target grade to include in the merger. For example, targeting grade 7 output with `ws@1` results in merging adapters for grades $\{5,6,7\}$, and `ws@1` results in merging adapters for grades $\{4,5,6,7,8\}$. Due to the nature of simplification grades and the limitations of our dataset, the upper and lower bounds of grade 12 and grade 2 remain as a boundary on the merge window. For example, target grade 3 with `ws@5` results in merging adapters for grades $\{2,3,4,5,6,7,8\}$, as there are no adapters for grades $\{-2,-1,0,1,2\}$, and targeting grade 12 with `ws@2` results in merging adapters for grades $\{10,11,12\}$, as there are no grade adapters above grade 12. As an additional control over number of merged adapters, we also perform “`n-copies`” experiments, where $n \in \{1, \dots, 11\}$ iterations of the same target grade adapter are merged together.

Weight Aggregation and Scaling. We begin testing to determine best practices for determining the scale of weighted adapters. We initially explore variations where the weights of all adapters sum to 1, where the weight magnitudes depend on the number of merged adapters, and compare these with weights averaging to 1. We use weighting (`w@`) to denote the specific weighting scheme used during adapter merger. As described above, we use three primary weighting strategies: uniform, proximity, and random. These strategies are further specified by a scaling factor:

For uniform weighting schemes, we investigate the effect that weight magnitude has on merge performance. We scale all weights (w) in a uniform merger by a linear scaling factor $\{0.5w, 1.0w, 2.0w, 3.0w\}$ to establish best practices for merge weight magnitudes.

For the proximity weighting functions (i.e., inverse distance) we test whether sharper or smoother weighting curves improve simplification results by scaling the weights peak by some geometric factor $\{d^{1/2}, d^1, d^2, d^3\}$, where $\frac{1}{d}$ is the distance from the target grade. The sharper the peak, the more similar it should be to single-adapter performance, while a flatter peak should more closely approximate a uniform weight distribution.

As an additional experimental variable and baseline for comparison, we also run model mergers where adapter weights are assigned randomly, with average magnitudes at $\{1.0, 2.0, 3.0\}$.

Adapter Selection. Two configurations introduce deliberate grade shifts (e.g., **upshifted-n**, **downshifted-n**) to assess systematic bias in readability control. These experiments probe whether merged adapters tend to overshoot or undershoot the prompted grade level, and if this can be compensated for by raising or lowering the grade levels of the merged adapters with respect to target simplification grade.

We also investigate whether merging only adapters *surrounding but excluding* the target grade are able to approximate graded output for the missing target grade. For example, target grade 6 with **no-tgt-grade** merging and **ws@2** results in merged adapters for grades $\{4, 5, 7, 8\}$.

Shared Warm-up vs. OTS. Finally, we compare merged adapters trained from the shared warm-up model against merged adapters derived from adapters trained directly on the OTS model (prefixed with **ots_**, as per our naming convention detailed above). This comparison evaluates whether shared initialization improves merger stability and reduces performance degradation due to parameter interference.

CHAPTER 5

EVALUATION FRAMEWORK

Evaluating TS systems requires balancing several partially competing objectives. A successful simplification must reduce linguistic complexity while preserving the meaning of the original text and maintaining both grammaticality and fluency. In the context of L2 education, we have an additional requirement: the resulting text should align with a specific target readability level given a learner’s proficiency. As no single metric captures all of these dimensions, evaluation in TS typically relies on a combination of complementary measures.

This chapter outlines the evaluation framework we use to assess the models developed in this thesis. The primary focus of the evaluation is readability alignment, measured using ΔFKGL , which serves as the primary metric for determining whether generated outputs match the intended grade level, and if so, how closely. To ensure that simplification does not degrade meaning or linguistic quality, the evaluation also incorporates standard simplification and semantic similarity metrics; namely SARI and BERTScore F1. Together, these measures provide a more balanced assessment of output readability, simplification behavior, and semantic preservation.

The following sections describe each of these metrics in detail, as well as explaining how they are applied to evaluate the effectiveness of the grade-specific adapters and adapter-merging strategies.

5.1 Fluency Evaluation (Perplexity)

We use perplexity as a proxy measure for evaluating the language model’s ability to generalize to unseen data. In this context, perplexity quantifies how well the finetuned model predicts the next token in a sequence, with lower perplexity scores indicating higher predictive confidence and, indirectly, better model fit. Because TS requires both semantic preservation and controlled complexity reduction, perplexity provides a useful baseline for assessing fluency and grammatical well-formedness of the generated output (Klakow and Peters, 2002).

For this experiment, perplexity is computed on a held-out test set consisting of unseen simplified texts from the dataset described in Section 3.2. This assesses how well the model captures the properties of simplified text at the target level. While we will also report scores of our trained models’ generated outputs, perplexity on held-out simplifications from our test sets offers a useful proxy for how effectively the model has internalized the simplification objective, and whether it generalizes beyond the specific examples seen during training. While perplexity alone does not capture the adequacy or appropriateness of a given simplification, it serves as an important sanity check for overall model performance.

5.2 Readability (FKGL)

To quantify the simplification achieved by the finetuned models, this study relies primarily on the FKGL (Kincaid, 1975). FKGL estimates reading difficulty via a function combining sentence length and average syllables per word into a score corresponding to U.S. school grade levels, making it transparently interpretable for the fine-grained grade-specific simplification approach pursued in this work.

The FKGL score is computed as:

$$FKGL = 0.39\left(\frac{totalwords}{totalsentences}\right) + 11.8\left(\frac{totalsyllables}{totalwords}\right) - 15.59$$

This work uses FKGL due to its simplicity, accessibility, and direct alignment with educational grade levels, which closely matches the level-specific simplification objectives of this research. Although FKGL does not account for all dimensions of text difficulty experienced by L2 learners, it offers a transparent and replicable baseline for assessing whether the generated outputs match the intended target complexities. In this study, we compute FKGL scores using the Python `readability` package, which provides standardized implementations of common formula-based readability metrics.

5.3 Simplification Quality (SARI, BERTScore F1)

While $\Delta FKGL$ serves as the primary optimization target in this study, readability level alignment alone does not ensure that a generated output is also a high-quality simplification. A model may reduce sentence length or syllable count while distorting meaning or arbitrarily removing essential information. To safeguard against this kind of improper simplification, we evaluate simplification quality using two complementary metrics: SARI and BERTScore F1.

SARI As introduced in Chapter 2, SARI evaluates the quality of simplification by measuring how well a model performs three primary operations: additions, deletions, and word retention relative to the source sentence and one or more human-written references (Xu et al., 2016). Unlike BLEU, which rewards n-gram overlap, SARI is designed specifically for text simplification and penalizes both excessive deletion and insufficient modification.

By definition, SARI computes F1 scores over three components:

- **Add:** rewarding appropriate lexical insertions (words present in the predicted and target simplifications, which were not present in the source sentence)
- **Delete:** rewarding appropriate removal of complex or unnecessary content (words that are present in the source sentence but not present in the predicted or target simplifications)
- **Keep:** rewarding preservation of important source content (words that are present in the source, target, and predicted sentences)

The final SARI score is the arithmetic mean of these three sub-scores.

We calculate SARI as the arithmetic mean of the F1 scores for the three edit operations:

$$\text{SARI} = \frac{F1_{\text{add}} + F1_{\text{keep}} + F1_{\text{delete}}}{3}$$

where each component F1 score is defined as:

$$F1 = 2 \cdot \frac{P \cdot R}{P + R}$$

and P and R denote precision and recall for the respective operation.

In this study, SARI scores are computed between the generated simplifications, the original source sentences, and the corresponding human reference simplifications from the dataset, as described in Section 3.2. We use the standard implementation provided in the HuggingFace `evaluate` library to ensure comparability with prior and future work.

In this research, SARI serves two primary purposes. First, it ensures that improvements in grade-level alignment are not achieved by trivial shortening or excessive deletion. Second, it provides an interpretable signal regarding whether merged adapters preserve appropriate simplification behavior relative to human references.

However, SARI does not directly measure semantic similarity beyond surface-level n-gram comparisons. For this reason, it is paired with a semantic similarity metric.

BERTScore We use BERTScore F1 to evaluate the semantic similarity between generated text and reference text using contextualized embeddings derived from pretrained transformer models (Zhang et al., 2020). Rather than relying on surface n-gram overlap, BERTScore F1 aligns tokens in embedding space and computes precision, recall, and F1 scores based on cosine similarity.

In this study, BERTScore F1 is computed between each generated simplification and its corresponding human reference simplification using the default `roberta-large` model. We report BERTScore F1 as our primary semantic similarity measure. For our purposes, there are two important ways in which BERTScore F1 complements SARI. First, it captures meaning preservation when lexical substitutions (i.e., simplifications) differ from the reference text. Second, it penalizes semantic drift that may not be detectable through n-gram-based metrics.

Because ΔFKGL remains our primary optimization objective, BERTScore F1 functions as a safeguard against unintended semantic degradation. In particular, during adapter merging experiments (see Section 4.5), BERTScore F1 allows us to verify that weighted parameter models do not produce uncontrolled shifts in meaning while optimizing for ΔFKGL .

5.4 Metric Interactions and Evaluation Strategy

No one metric fully captures simplification quality. ΔFKGL measures structural readability, SARI measures edit-based simplification distance, and BERTScore F1 measures semantic preservation. These metrics are therefore considered important indications of simplification adequacy.

Across experiments, we prioritize configurations that minimize ΔFKGL from the requested target grade, maintain or improve SARI relative to baseline systems, preserve high BERTScore F1 values to ensure semantic adequacy, and avoid increases in perplexity indicating degraded fluency. This multi-metric evaluation framework ensures that improvements in readability control are not achieved at the expense of meaning preservation or grammaticality.

CHAPTER 6

RESULTS

In this chapter, we present our evaluation of the controllable TS system and examine the three primary RQs, as detailed in chapter 1.2, namely: (1) whether grade-level simplification can be captured through adapter fine-tuning, (2) whether adapter merging results in increased control and stability, and (3) whether weighted combinations of fine-tuned adapters allow for controllable graded output along the readability spectrum.

As outlined in chapters 2.4 and 5, model outputs are evaluated using three automatic metrics:

- **ΔFKGL** , which measures the distance between target grade and readability grade of the generated output.
- **SARI**, which measures the quality of the simplification by counting edits relative to the source and target.
- **BERTScore F1**, which measures the simplification adequacy, by means of comparing semantic similarity between predictions and target simplifications.

We determine success and failure using ΔFKGL , while relying on our two secondary metrics, SARI and BERTScore F1 to catch model deviations not accounted for by changes in sentence structure.

In the following sections, we organize our results according to the model’s experimental stages, which reflect the increasing complexity of the system architecture. We first compare baseline models against discrete adapters, then systematically vary adapter-merging configurations and parameters to analyze how these permutations affect readability control and simplification quality.

6.1 Baseline System

We begin by establishing a baseline against which we can draw future comparisons, as well as motivate the research itself. To do this, we measure the performance of an unmodified LLaMA3.2-3B-Instruct model and take the average performance across all graded datasets. From these results, we can make several observations:

Run Name	Δ FKGL	SARI	BERTScore F1
off-the-shelf baseline	3.53	33.62	87.04

TABLE 6.1: Baseline model performance across evaluation metrics.

We see immediately that the OTS model performs poorly by all metrics, as suggested by preliminary results prior to the main research phase. For the perplexity score on the training dataset, it’s over 9,000! When compared to scores of 30-50 for shared warm-up and graded-from-off-the-shelf models, respectively, this indicates a severe mismatch between the OTS model’s probability distribution and that of the dataset. With a mean Δ FKGL of 3.53, the OTS model consistently produces content far above the targeted readability level, and a SARI score of 33.62 indicates that very little of the content from the source sentence or target simplification make it into the predicted generation. This finding is unsurprising, as we assume the model has no prior knowledge of grade-based simplifications, and no structural knowledge about readability grade levels.

6.2 Shared Warm-up Model

The shared warm-up model represents the first stage of task-specific learning. Rather than beginning with individual graded adapters, we start by fine-tuning a single adapter on the full combined dataset for all grades to serve as a shared initialization point. This serves two purposes: First, we expose the model to the general simplification task, where it begins to learn the structural relationship between source and simplified output sentences. Second, the shared initialization point from which all future grade-specific adapters are trained reduces parameter conflicts during later mergers, since all adapters share common knowledge of the full dataset.

Because training data covers a range of grade levels, the shared warm-up model is not specialized to any particular grade, instead learning a generalized simplification behavior reflecting the broader distribution within our dataset. Evaluating this model therefore provides initial insights into what we can expect in terms of model performance during later fine-tuning stages and merger, where adapters become increasingly specialized.

The results in table 6.2 (below) show that even generalized finetuning produces substantial task-specific improvements over OTS performance. The mean Δ FKGL drops from 3.53 to 1.18, indicating that our generalized adapter is capable of producing simplifications much more closely aligned with the target sentences at the intended readability level. This is a strong first indication that the notion of grade-specific simplification can be inferred through examples, without explicit control over the (linguistic) parameters involved in calculating readability level.

Run Name	Δ FKGL	SARI	BERTScore F1
off-the-shelf baseline	3.53	33.62	87.04
shared warm-up	1.18	75.88	95.92

TABLE 6.2: Shared warm-up vs baseline model performance.

The increase of the SARI score from 33.62 to 75.88 is our strongest indication that the fine-tuned model is beginning to produce relevant and appropriate output. At the same time, a BERTScore F1 of 95.92 suggests that the shared warm-up model also more effectively preserves the semantic component for these simplification operations.

While the Δ FKGL score remains higher (worse) than those achieved by the grade-specific adapters introduced in the following section, this is an expected limitation of the shared warm-up stage: a model covering all of the data with equal importance learns to effectively mimic our simplification task in a broad sense, at the same time, it lacks the specificity of adapters trained on single grade datasets.

6.3 Individually Trained Graded Adapters

In addition to the two baseline models, we evaluate the performance of grade-specific adapters trained for each target readability level. Each adapter is trained on the subset of the dataset corresponding to its target grade. We perform this training under two initialization conditions: adapters trained directly from the OTS base model, and adapters initialized from the shared warm-up model described in Section 6.2. These configurations produce four pre-merge baseline systems that will serve as reference points for the later adapter-merging experiments.

Run Name	Δ FKGL	SARI	BERTScore F1
off-the-shelf baseline	3.53	33.62	87.04
shared warm-up	1.18	75.88	95.92
graded-from-off-the-shelf	1.02	76.97	95.73
graded-from-shared-warmup	1.03	76.50	95.87

TABLE 6.3: Graded adapter performance vs baselines.

The result of this additional fine-tuning is a substantial performance improvement over both baseline systems. Among the evaluated configurations, the graded-from-off-the-shelf adapters achieve the lowest ΔFKGL and the highest SARI. However, the performance difference between the two graded configurations is minimal. The graded-from-shared-warmup adapters produce nearly identical scores across all metrics. Both graded configurations reduce ΔFKGL to approximately one grade level. This indicates that the models learn to generate outputs closely aligned with the requested readability level despite receiving no explicit definition of grade boundaries during training. These results are expected. Because each adapter is trained on a narrow subset of the dataset corresponding to a single grade level, the models specialize toward the linguistic patterns and simplification strategies present in that subset. As a result, the individual graded adapters represent the strongest task-specific models in this study.

However, the goal of this research is not simply the training of specialized adapters for every grade level. Our objective is to investigate whether adapter merger can provide stable and flexible control over readability output across the grade spectrum. For this reason, the graded adapter set-ups may serve as a useful upper-bound reference point which later merging experiments will attempt to meet and exceed.

These results confirm two important concepts: First, task-specific fine-tuning is essential for controllable sentence simplification, as OTS performance scores poorly. Second, specialized grade-level adapters perform very well, exceeding the ability of generalized fine-tuning to accurately align with grade-level simplifications. These serve as a meaningful benchmark against which our merged adapter configurations may be evaluated.

6.4 Merged Adapters

We next evaluate whether combining grade-specific adapters through parameter merging improves readability control. These experiments compare several merge operators and weighting schemes applied to the full set of graded adapters.

6.4.1 Adapter Merging Methods

In section 4.5.1 we described the evaluation of several adapter merging strategies implemented in the HF adapter merging framework via `add_weighted_adapter()`. These include linear merging, concatenation, TIES, DARE-TIES, DARE-linear, and magnitude pruning. Each method was applied to the full set of eleven graded adapters and evaluated across all target grade levels.

As we alluded to in chapter 3, nearly all merge operators resulted in catastrophic failure. Across linear, CAT, TIES, DARE-linear, and magnitude-pruning configurations, generated

outputs frequently collapsed into one or more of the following behaviors: non-language character sequences, verbatim copying of the source sentence, repetitive loops, or outputs whose readability diverged dramatically from the requested grade level. In these cases, ΔFKGL values commonly fell between 20 and 200, indicating a complete loss of control over readability.

Among the evaluated methods, only **DARE-TIES** consistently produced stable outputs. As a result, the remainder of the experiments in this study focus on DARE-TIES configurations. However, we report a small set of linear merging experiments as an additional baseline comparison, since linear merging represents the most widely used and conceptually straightforward adapter merging strategy. Table 6.4 summarizes the performance of the evaluated linear merge configurations.

Run Name	ΔFKGL	SARI	BERTScore F1
<i>Baselines</i>			
off-the-shelf baseline	3.53	33.62	87.04
shared warm-up	1.18	75.88	95.92
graded-from-off-the-shelf	1.02	76.97	95.73
graded-from-shared-warmup	1.03	76.50	95.87
<i>Linear Merge</i>			
linear_ws@n_w@proximity_average	40.86	18.20	64.58
linear_ws@n_w@proximity_sum	75.76	24.28	66.13
linear_ws@n_w@uniform_average	40.09	18.13	64.35
linear_ws@n_w@uniform_sum	69.17	23.30	65.79

TABLE 6.4: Performance of linear adapter merging configurations averaged across window sizes.

The results confirm the qualitative observations described above. Across all linear configurations, readability control deteriorates severely. Even the best-performing runs produce ΔFKGL values above 20, indicating a complete loss of control. In many cases, errors exceed 100 grade levels, reflecting total collapse of the merged adapter and complete failure to perform the intended behavior. Simplification quality metrics degrade similarly. SARI scores fall below 40 across all configurations, which is substantially lower than the scores achieved by both the shared warm-up and the graded adapter models introduced earlier. BERTScore F1 values also drop by more than 20 points relative to the baseline systems, indicating significant semantic drift or incoherent generation.

Taken together, these results demonstrate that naive linear merging introduces severe destructive interference between adapter parameters. When multiple grade-specific adapters are combined through simple weighted addition, their task-specific parameters appear to compete rather than cooperate, resulting in unstable and often unusable outputs. This outcome contrasts with prior findings ala. Adapter Soup/Fusion (Chronopoulou et al., 2023, Pfeiffer et al., 2021), where simple parameter averaging preserves or improves model performance when combining related task models.

For this reason, subsequent experiments focus exclusively on DARE_TIES merging, which proved to be the only method capable of combining multiple graded adapters while preserving readability control and producing coherent outputs.

6.4.2 Merge Permutations

While grade-specific adapters achieve strong performance, their primary limitation lies in the discrete grade levels on which they are trained. In practical settings, the sparsity of grade level appropriate data is limited, while readability requirements may simultaneously fall between discrete readability levels (i.e., grade 7.5). To address this limitation, our study explores adapter merging as a method for supplementing the low-resource nature of graded simplification data. In our experiments, multiple grade-specific adapters are merged together at inference time to produce a model where the behavior lies somewhere between the individual models' outputs.

To begin, we compare the performance of our baseline models, specifically the shared warm-up model and graded from OTS, with a merged adapter containing all graded adapters for grades 2 through 12. We use both uniform weight distributions (all adapters set to weight 1) as well as random weights using a Dirichlet distribution, where adapter weights average to 1, 2, and 3. As in all other experiments with weight scaling > 1 , random weights with averages at 2 and 3 result in catastrophic failure, and we do not group them with our baseline results (see Appendix A for the full results table).

Run Name	Δ FKGL	SARI	BERTScore F1
<i>Baselines</i>			
off-the-shelf baseline	3.53	33.62	87.04
shared warm-up	1.18	75.88	95.92
graded-from-off-the-shelf	1.02	76.97	95.73
graded-from-shared-warmup	1.03	76.50	95.87
<i>All-grades</i>			
all-grade merge (uniform weights)	1.12	76.31	95.90
all-grade merge (random weights)	1.17	74.95	95.72

TABLE 6.5: Baseline model performance vs naive merger of all adapters.

Table 6.5 shows the performance of our naive full merger. This naive full merger demonstrates that merging grade-specific adapters stabilizes the performance over the single adapter containing all grade data, while still not meeting the performance of the individual specialized adapters. Both uniform and random weights preserve BERTScore F1, while random weights degrade Δ FKGL and SARI scores when compared to uniform weights. In this trial, random weights result in the lowest SARI score (besides the OTS model), suggesting that adapter weights deviating from 1 begin to cause simplification degradation. We will consider this further in section 6.4.4 when discussing proximity peak steepness and adapter weight scaling.

6.4.3 Window Size

We next examine how the number of related adapters in the merger affects readability control. The window-size parameter - `ws@` - determines which grade adapters are included in the merged model around the target grade. A window size of 1 merges the target adapter with grades 1 higher and 1 lower, while larger windows include adapters from progressively more distant grade levels. Because grade boundaries truncate the merge window near the edges of the grade range (2 on the low end, 12 on the high end), the effective number of merged adapters varies depending on the target grade. We therefore consider both the listed `ws@` parameter and the resulting number of adapters included in the merger.

Run Name	ws@	Δ FKGL	SARI	BERTScore F1
<i>Baselines</i>				
off-the-shelf baseline		3.53	33.62	87.05
shared warm-up		1.18	75.88	95.92
graded-from-off-the-shelf		1.02	76.97	95.73
graded-from-shared-warmup		1.03	76.50	95.87
<i>Window Size Comparisons</i>				
dareties_w@doubled_average	ws@1	2.29	73.05	91.46
dareties_w@uniform_sum	ws@1	3.08	35.44	88.53
dareties_w@proximity_sum	ws@1	3.13	35.29	88.45
dareties_w@tripled_average	ws@1	18.84	29.52	23.05
ots_dareties_w@uniform_average	ws@2	1.05	76.88	95.72
ots_dareties_w@proximity_average	ws@2	1.06	76.61	95.76
dareties_w@uniform_average	ws@3	1.06	76.35	95.85
dareties_w@proximity_average	ws@3	1.07	76.24	95.78
dareties_w@proximity-squared_average	ws@3	1.15	74.85	95.68
dareties_w@proximity-flatter_average	ws@4	1.06	76.50	95.85
dareties_w@halved_average	ws@5	1.86	53.88	94.82

TABLE 6.6: Best performing window size per weighting configuration according to Δ FKGL (lower is better), alongside SARI and BERTScore F1.

Across the configurations that do not result in catastrophic failure (i.e., Δ FKGL > 5), increasing the merge window from `ws@1` to `ws@2` consistently reduces the deviation from the target grade. For example, under uniform averaging, the mean deviation decreases from Δ FKGL = 1.38 at `ws@1` to 1.08 at `ws@2`. Further increases in window size produce only small changes: `ws@3` yields Δ FKGL = 1.06, while `ws@4` yields Δ FKGL = 1.07. A similar pattern appears with the default proximity_average experiments, where Δ FKGL decreases from 1.19 at `ws@1` to 1.08 at `ws@2`, and remains near 1.06-1.08 for `ws@3-4`. The other proximity-based weighting schemes show comparable behavior. For example, proximity-flatter improves from Δ FKGL = 1.19 at `ws@1` to 1.07 at `ws@2`, and remains within 1.06-1.07 for `ws@3-5`. Proximity-squared averaging shows a slightly higher but similarly stable range, improving from 1.26 at `ws@1` to 1.15 at `ws@2`, and remaining near 1.15-1.21 for `ws@3-5`, while proximity-cubed degrades further, showing stabilization going from `ws@1` to `ws@2`, but increased instability beyond `ws@3`.

For “successful” runs with larger windows, the ΔFKGL values remain relatively stable across most averaging configurations. For example, uniform averaging remains within 1.06-1.09 for $\text{ws}@3-5$, and the corresponding “n-copies” experiments, which merge repeated instances of the same adapter and therefore isolate the effect of increasing the number of merged adapters without adding further context or capabilities, produce values between 1.03 and 1.11 across $\text{ws}@3-5$.

The OTS adapter experiments show the same trend. Under OTS_dareties with uniform averaging, ΔFKGL improves from 1.18 at $\text{ws}@1$ to 1.05 at $\text{ws}@$, and remains within 1.05-1.12 for $\text{ws}@3-5$.

In contrast, weighting schemes that scale adapter contributions, by averaging to 2 or 3 rather than averaging them to 1, show substantially larger deviations from the target grade. For example, doubling all adapter weights yields $\Delta\text{FKGL} = 12.54$ at $\text{ws}@2$, and tripling produces $\Delta\text{FKGL} = 24.06$ under the same window size. Linear-weighting variants produce even larger deviations, with all ΔFKGL values exceeding 20 and reaching over 100 in some configurations.

Overall, the results show a consistent reduction in ΔFKGL when increasing the merge window from $\text{ws}@1$ to $\text{ws}@2$, after which larger windows or larger numbers of merged adapters either produce only small fluctuations in the deviation from the target grade or cause rapid degradation in mergers with destructive weighting schemes (e.g., scaling, sum, linear).

6.4.4 Weight Aggregation and Scaling

In addition to varying the merge window size, we evaluate how different weight aggregation strategies influence the merged adapter performance. These experiments modify how adapter weights are normalized or scaled before merging. We consider three main groups of strategies: (1) uniform weight distributions with linear scaling, (2) inverse distance (proximity) weighting with geometric scaling, and (3) random weights with scaled averages. Table 6.8 reports mean performance across all window sizes for each weighting scheme.

Uniform Weighting. Uniform weighting assigns equal weights to all adapters within the merge window, trials use weights in $\{0.5w, 1.0w, 2.0w, 3.0w\}$. The standard averaging ($1.0w$) configuration produces stable results, with mean $\Delta\text{FKGL} = 1.13$. Adjusting the magnitude of these weights quickly degrades performance. Halving the weights increases the deviation to $\Delta\text{FKGL} = 2.09$. Increasing the magnitude produces much larger deviations: doubled weights yield $\Delta\text{FKGL} = 13.82$ and tripled weights reach $\Delta\text{FKGL} = 27.82$. The pattern is consistent across window sizes.

Proximity Weighting. Proximity weighting assigns weights proportional to the inverse distance ($\frac{1}{d}$) from the target grade, with scaling factors in $\{d^{1/2}, d^1, d^2, d^3\}$. The baseline configuration (d^1) produces mean $\Delta\text{FKGL} = 1.10$. Flattening the weighting curve slightly improves performance, with the proximity-flatter variant ($d^{1/2}$) achieving $\Delta\text{FKGL} = 1.09$, the lowest average deviation among the weighting strategies tested. Increasing the steepness of the weighting curve has the opposite effect. Squaring the weights increases the deviation to $\Delta\text{FKGL} = 1.19$, while cubing them produces $\Delta\text{FKGL} = 1.50$. These changes reflect how strongly the weighting distribution concentrates around the central adapter: flatter distributions behave similarly to uniform averaging, while sharper peaks degrade performance by including weights with extreme magnitudes in either direction.

Although the mean ΔFKGL values for the standard proximity and uniform weighting are similar (1.10 vs. 1.13, respectively), we also compare the methods on a per-grade basis. For each weighting strategy (proximity vs uniform), across all window sizes ($\{1, 2, 3, 4, 5\}$) and target grades ($\{2, \dots, 12\}$) we record which weighting strategy produces the smaller ΔFKGL . Under shared warm-up initialization, proximity weighting produces lower ΔFKGL values in 60% of cases, and 47% for the OTS initialized adapters.

Comparison	Dominant Method	Ratio
Proximity vs. Uniform (Shared Warm-up)	Proximity	60%
Proximity vs. Uniform (Off-the-Shelf)	Uniform	53%

TABLE 6.7: Per-grade dominance ratios comparing weighting strategies and initialization methods. For each target grade, the configuration producing the lower ΔFKGL is counted as dominant.

Random Weighting. Random weighting assigns adapter weights sampled from a Dirichlet distribution which guarantees that their average magnitude matches a specified value in $\{1.0, 2.0, 3.0\}$. When the average magnitude is near 1.0, the resulting merges remain relatively stable, producing mean $\Delta\text{FKGL} = 1.17$. Increasing the magnitude quickly destabilizes the merge. At magnitude 2.0 the deviation rises to $\Delta\text{FKGL} = 5.96$, and at magnitude 3.0 it increases further to $\Delta\text{FKGL} = 20.23$. These results illustrate that arbitrary weighting distributions can occasionally approximate the behavior of structured weighting schemes when the scale is constrained, but they become unreliable as the magnitude of the weights increases.

Weight Scheme	Mean Δ FKGL	Mean SARI	Mean BERTScore F1
<i>Baselines</i>			
off-the-shelf baseline	3.53	33.62	87.05
shared warm-up	1.18	75.88	95.92
graded-from-off-the-shelf	1.02	76.97	95.73
graded-from-shared-warmup	1.03	76.50	95.87
<i>Uniform Weights</i>			
dareties_w@halved_average	2.09	49.47	93.67
dareties_w@uniform_average	1.13	75.28	95.70
dareties_w@doubled_average	13.82	43.32	42.06
dareties_w@tripled_average	27.82	34.89	48.71
<i>Proximity Weighting</i>			
dareties_w@proximity-flatter_avg	1.09	75.61	95.80
dareties_w@proximity_avg	1.10	75.47	95.77
dareties_w@proximity-squared_avg	1.19	73.71	95.68
dareties_w@proximity-cubed_avg	1.50	64.20	95.49
<i>Random Weighting</i>			
dareties_w@random-1_average	1.17	74.95	95.72
dareties_w@random-2_average	5.96	63.60	77.61
dareties_w@random-3_average	20.23	25.32	15.79

TABLE 6.8: Average performance across all window sizes for weight scaling strategies. Lower Δ FKGL indicates smaller deviation from the target grade.

6.4.5 Adapter Selection

To evaluate whether systematic bias appears in the merged adapters (i.e., consistently generating above or below the target grade level), we test several configurations that deliberately shift the set of merged grades relative to the target grade. These experiments adjust the merge window upward or downward by one or two grade levels to measure whether the resulting simplifications consistently over- or undershoot the desired readability level. We also test configurations where the target-grade adapter itself is excluded from the adapter merger.

Downshifted Adapter Selection. Downshifted configurations merge adapters from grades below the requested target level. When the merge window is shifted down by one grade, the resulting simplifications produce mean Δ FKGL = 1.10. Shifting the window down by two grades increases the deviation slightly to Δ FKGL = 1.12. These values remain close to the deviations observed under the baseline uniform weighting configurations.

Upshifted Adapter Selection. Upshifted configurations merge adapters from grades above the requested target level. A one-grade upward shift produces mean Δ FKGL = 1.10, which closely matches the downshifted-1 configuration. Increasing the shift to two grades results in Δ FKGL = 1.12. Across both upshifted configurations, the deviations from the target grade

remain within a narrow range comparable to the baseline averaging results. Note: these results coincidentally match the downshifted results almost exactly, only deviating after the third or fourth decimal, in spite of showing entirely different trend lines, as seen in figure A.1. We take this as a surprising happenstance, and do not infer any structural alignment from the two experiments.

Excluding the Target Adapter. Finally, we evaluate whether the merged adapters surrounding a target grade can approximate the target readability level when the corresponding adapter is excluded from the merge. In these experiments, the adapter set includes only grades surrounding the target grade. This configuration produces mean $\Delta\text{FKGL} = 1.15$. While this deviation is slightly larger than the shifted configurations above, it remains within the same general range observed across the more successful merges.

Run Name Prefix	Mean FKGL	Mean SARI	Mean BERTScore F1
<i>Baselines</i>			
off-the-shelf baseline	3.53	33.62	87.05
shared warm-up	1.18	75.88	95.92
graded-from-off-the-shelf	1.02	76.97	95.73
graded-from-shared-warmup	1.03	76.50	95.87
<i>Downshifted Adapter Selection</i>			
dareties_w@uniform_downshifted-1	1.10	75.36	95.77
dareties_w@uniform_downshifted-2	1.12	75.05	95.73
<i>Upshifted Adapter Selection</i>			
dareties_w@uniform_upshifted-1	1.10	76.00	95.86
dareties_w@uniform_upshifted-2	1.12	75.78	95.89
<i>Target Adapter Excluded</i>			
dareties_w@uniform_no-tgt-grade	1.15	74.18	95.81

TABLE 6.9: Average performance across all window sizes for adapter selection experiments. Downshifted and upshifted configurations shift the merged adapter set relative to the target grade, while the no-tgt-grade configuration excludes the target-grade adapter entirely.

6.4.6 Shared Warm-up vs. OTS

The final experiment compares adapter merging using adapters initialized from the shared warm-up model with adapters trained directly from the OTS base model. These configurations test whether initializing adapters from a shared warm-up stage produces more consistent readability control when the adapters are later merged. Table 6.10 reports the mean performance of each configuration across all window sizes.

Weight Scheme	Mean Δ FKGL	Mean SARI	Mean BERTScore F1
<i>Baselines</i>			
off-the-shelf baseline	3.53	33.62	87.05
shared warm-up	1.18	75.88	95.92
graded-from-off-the-shelf	1.02	76.97	95.73
graded-from-shared-warmup	1.03	76.50	95.87
<i>Shared Warm-up Initialization</i>			
dareties_w@proximity_average	1.10	75.47	95.77
dareties_w@uniform_average	1.13	75.28	95.70
<i>Off-the-Shelf Initialization</i>			
ots_dareties_w@uniform_average	1.10	76.13	95.72
ots_dareties_w@proximity_average	1.10	76.02	95.74

TABLE 6.10: Average performance across all window sizes for merges using shared warm-up adapters and OTS adapters.

Shared Warm-up. Adapters trained from the shared warm-up model produce mean Δ FKGL = 1.10 under proximity weighting and Δ FKGL = 1.13 under uniform weighting. These values fall within the range observed across the stable averaging configurations discussed in earlier experiments. The proximity-weighted merges produce slightly smaller deviations from the target grade than uniform weighting.

Off-the-Shelf. Adapters trained directly from the OTS model produce nearly identical deviations from the target grade. Under both uniform and proximity-based weighting, the merged adapters produce mean Δ FKGL = 1.10. These values closely match the results obtained from the shared warm-up adapters, and across both initialization strategies, the merged adapters remain within a narrow range of deviations from the target grade. The primary difference appears in the secondary metrics, where the OTS configurations produce slightly higher SARI scores while maintaining comparable BERTScore F1 values. Because the mean performance of the two initialization strategies is nearly identical, we also compare them on a per-grade basis. For each weighting strategy ($\{\text{proximity, uniform}\}$), across all window sizes ($\{1, 2, 3, 4, 5\}$) and target grades ($\{2, \dots, 12\}$) we record which initialization produces the lower Δ FKGL. In 55% of these cases, the shared warm-up adapters outperform the OTS adapters. Table 6.11 summarizes these comparisons for proximity versus uniform weighting under both initialization strategies, as well as the overall comparison between OTS and shared warm-up adapters.

Comparison	Dominant Method	Ratio
Proximity vs. Uniform (Shared Warm-up)	Proximity	60%
Proximity vs. Uniform (Off-the-Shelf)	Uniform	53%
Off-the-Shelf vs. Shared Warm-up	Shared Warm-up	55%

TABLE 6.11: Per-grade dominance ratios comparing weighting strategies and initialization methods. For each target grade, the configuration producing the lower Δ FKGL is counted as dominant.

CHAPTER 7

DISCUSSION

This chapter interprets the experimental findings presented in Chapter 6 and evaluates how they address the research questions introduced in Chapter 1. The results demonstrate that adapter merging can provide reliable control over readability levels while preserving simplification quality across a range of merging configurations. At the same time, the experiments reveal several constraints on how adapter mixtures behave, particularly with respect to weight scaling, merge window size, and adapter initialization.

We begin by interpreting the main empirical patterns observed in the experiments. We then analyze the findings with respect to the three research questions guiding this work. Finally, we compare the results to prior work on controllable generation and adapter composition, discuss implications for second language learning applications, and outline the limitations of the present study.

7.1 Interpretation of Findings

Taken together, the experiments suggest that the behavior of merged adapters is governed by three interacting factors: the scale of the merge weights, the density of the merged representation, and the redundancy between adjacent grade adapters. The results do not indicate that adapter merging works reliably under arbitrary conditions. Rather, stable readability control appears only within a constrained region of the merge space where these factors remain balanced.

Weight Scale Stability. The clearest pattern across the experiments is that merge stability depends heavily on preserving the relative magnitude of adapter weights. Configurations that remain close to normalized averaging consistently produce the smallest deviations from the requested grade. For example, proximity and uniform weighting averaged to 1 both remain near $\Delta\text{FKGL} = 1.10$ on average, while proximity-flatter weighting produces $\Delta\text{FKGL} = 1.09$.

In contrast, altering the scale of these weights quickly destabilizes the merge. Halving the weights increases the deviation to $\Delta\text{FKGL} = 2.09$, while doubling them increases it to $\Delta\text{FKGL} = 13.82$ and tripling them to $\Delta\text{FKGL} = 27.82$. The same trend appears in the proximity scaling experiments: squaring the scaling factor raises the deviation to $\Delta\text{FKGL} = 1.19$ and cubing it increases the deviation to $\Delta\text{FKGL} = 1.50$. These results suggest that adapter merging functions as a controlled interpolation only when the merge weights remain close to their original normalized scale.

Merge Density and Parameter Sparsification. The window-size and n-copies experiments point to a second mechanism related to merge density. Under the DARE-based merge operator, portions of each adapter’s parameters are randomly dropped before merger. When only a small number of adapters are merged, these dropped parameters create sparse representations that may omit portions of the learned transformation, like taking ice cream scoops out of your brain before speaking. As additional adapters are included in the merge, the probability that a given parameter is present in at least one adapter increases. In effect, larger merges reconstruct a denser approximation of the underlying adapter(s).

This mechanism provides a plausible explanation for the plateau observed in the window-size experiments. Expanding the merge from `ws@@1` to `ws@@2` improves performance, but larger windows rarely produce further gains. The n-copies experiments support this interpretation from a different perspective: repeatedly merging copies of the same adapter quickly converges to stable ΔFKGL values once several copies are included. In both cases, the additional adapters appear to compensate for sparse parameters rather than contributing new information about the target grade.

Local Redundancy Among Grade Adapters. The adapter selection experiments reveal a third structural property of the trained adapters. Shifting the merge window upward or downward by one or two grade levels produces only small changes in performance, with mean deviations remaining between $\Delta\text{FKGL} = 1.10$ and $\Delta\text{FKGL} = 1.12$. Even excluding the target adapter entirely results in $\Delta\text{FKGL} = 1.15$. These results suggest that adjacent grade adapters encode overlapping simplification behavior. However, the evidence supports only a local form of redundancy rather than a globally smooth representation of grade levels. The dominance comparisons illustrate this point: proximity weighting produces lower ΔFKGL values than uniform weighting in 60% of grades under shared warm-up initialization, but only 47% under OTS initialization. Similarly, OTS adapters outperform shared warm-up adapters in only 45% of grades overall. These results indicate that differences between configurations are often small and inconsistent across individual grade levels.

Taken together, these findings suggest that grade-specific adapters share overlapping parameters for simplification, particularly among neighboring grade levels. The implication is that the simplification process for any given grade is largely similar when compared to that of the neighboring grades. This makes intuitive sense, but our findings serve to confirm this intuition. Outside of the adjacent grades’ neighborhoods, merge stability depends primarily on maintaining appropriate weight scaling and merge density, rather than properties of each grade’s simplification process itself.

7.2 Analysis of Research Question 1

Research Question 1 asks whether adapters trained on discrete grade levels produce outputs that reliably match their intended readability targets.

The baseline experiments indicate that grade-specific adapters do capture systematic differences in readability. When evaluated individually, both shared warm-up and OTS adapters produce outputs that remain close to their target grade levels, with deviations typically near $\Delta\text{FKGL} = 1.02$ to 1.18 depending on initialization. These values indicate that the adapters encode meaningful distinctions between the training grade levels rather than collapsing toward a single simplification strategy. At the same time, the magnitude of these deviations suggests that individual grade adapters do not produce perfectly calibrated outputs. A deviation of approximately one grade level implies that the adapters capture broad differences in complexity but do not map precisely onto the discrete grade boundaries used during training. In other words, the adapters appear to learn a vague association for appropriate simplification rather than a sharply separated grade-specific behavior.

This interpretation is reinforced by the adapter selection experiments. When the target adapter is removed from the merge entirely, surrounding grade adapters are still able to approximate the requested readability level with mean $\Delta\text{FKGL} = 1.15$. Similarly, shifting the merge window upward or downward by one or two grades changes performance only slightly. These results suggest that adjacent grade adapters encode overlapping simplification behavior rather than distinct, isolated representations, but the difference is not simply a systematic overshooting of the target grade (e.g., always producing results 1 grade level higher or lower than the target grade).

Single-grade adapters, therefore, provide only a coarse mechanism for readability control. Although they capture consistent differences in linguistic complexity corresponding to the grade levels present in the training data, the resulting outputs remain substantially noisier than the grading scheme would ideally allow. Even the strongest fine-tuned adapters deviate from their intended targets by roughly one grade level on average, whereas the grading scheme itself would

suggest an achievable precision closer to 0.25-0.5 grades. In practice, the adapters encode relative ordering between grade levels more reliably than they reproduce the precise grade boundary associated with each level.

7.3 Analysis of Research Question 2

Research Question 2 asks whether merging grade-specific adapters improves the consistency of readability control compared to single-adapter generation.

The results indicate that merging can approximate the performance of individually trained grade adapters, but it does not surpass the strongest fine-tuned baselines. The best single-adapter configuration, graded-from-off-the-shelf initialization, achieves $\Delta\text{FKGL} = 1.02$. The strongest merged configurations cluster slightly above this range, around $\Delta\text{FKGL} = 1.05\text{-}1.07$ under normalized uniform and proximity weighting. This outcome is not unexpected. The fine-tuned adapters represent the most direct optimization of the model parameters for the target readability level. In principle, they should therefore provide the best possible match to the training objective. One motivation for the merging experiments was the possibility that individually fine-tuned adapters might exhibit systematic bias, for example consistently overshooting or undershooting their target grade. If such bias were present, combining neighboring adapters through weighted merging might correct it by shifting the output toward the grades included in the merger. However, the experiments do not provide evidence for such systematic bias. The shifted-adapter experiments show only small changes in deviation, and the dominance comparisons do not reveal a consistent advantage for merged configurations. In practice, the individually trained adapters already appear to capture the intended grade simplification as well as can be achieved. As a result, merging primarily provides compositional flexibility rather than improved absolute accuracy.

The experiments also demonstrate that adapter merging is highly sensitive to how the merge is performed. Many configurations diverge substantially from the target grade. In particular, altering the scale of the merge weights produces large deviations from the requested readability level. Doubling the merge weights increases the average deviation to $\Delta\text{FKGL} = 13.82$, and tripling them increases it further to $\Delta\text{FKGL} = 27.82$. Similar patterns appear in the proximity scaling experiments, where increasingly sharp weighting curves gradually degrade performance. These results show that merging does not automatically produce stable readability control; stability depends on preserving a balanced weighting scheme during merger. The window-size and n-copies experiments further clarify the role of merging. Expanding the merge from a single adapter to a small neighborhood typically improves performance, but larger merges provide little additional benefit. This plateau appears once the merge includes several contributing adapters, suggesting that the primary effect of larger merges is to reconstruct a denser representation

after the sparsity introduced by the merge operator rather than to introduce additional grade information.

All in all, adapter merging does not resolve the primary limitation presented in RQ1. While merged adapters can reproduce the behavior of individually trained grade adapters and occasionally reduce small deviations, they do not outperform the strongest fine-tuned baselines. The best merged configurations remain clustered slightly above the single-adapter results, and many alternative merge settings diverge substantially from the target grade. As a result, merging primarily offers a mechanism for combining existing grade representations rather than improving the underlying accuracy of readability control.

7.4 Analysis of Research Question 3

Research Question 3 asks whether a weighting scheme over grade-specific adapters can generate text at intermediate readability levels that were not explicitly represented in the training data.

The experimental results provide evidence that weighted adapter mixtures can approximate nearby grade levels even when the exact target adapter is not included in the merge. In the target-exclusion experiment, surrounding grade adapters alone approximate the missing grade with mean $\Delta\text{FKGL} = 1.15$. Similarly, shifting the merge window upward or downward by one or two grades produces only small changes in deviation, with mean ΔFKGL remaining between 1.10 and 1.12. These results indicate that nearby grade adapters encode overlapping simplification behavior that can be combined to reconstruct adjacent readability levels. At the same time, the experiments do not support the stronger interpretation implied by the original hypothesis that a weighting scheme could function as a continuous “dial” across the entire readability spectrum. The dominance comparisons show that small differences between weighting strategies vary inconsistently across individual grades. For example, proximity weighting produces lower ΔFKGL values than uniform weighting in only 60% of tests under shared warm-up initialization but only 47% under OTS initialization. These results suggest that the interpolation behavior is not strongly encoded in the grade-specific adapters.

The experiments show that merging nearby adapters can reconstruct missing or intermediate grade levels, but only within a limited range of accuracy. In most cases, the resulting outputs remain within roughly 1-1.5 grade levels of the target rather than matching it precisely. The method therefore behaves more like a coarse interpolation between adjacent grade adapters than a continuous control mechanism across the readability spectrum. While this capability allows intermediate grades to be approximated without retraining additional adapters, it falls short of enabling the precise, dial-like control over readability that originally motivated the approach.

7.5 Comparison to Prior Work

This work sits between two lines of prior research that do not usually meet in the same system. On one side, recent controllable simplification work has focused on explicit control over the linguistic properties of the output. On the other, PEFT work has emphasized modularity and post-hoc composition of learned components. The present thesis borrows the modularity of adapters but applies it to a problem that more closely resembles controllable simplification: targeting output difficulty at specific grade levels.

The most direct comparison is to work such as Thillainathan and Koller (2024), which controls simplification through explicit linguistic targets, including dependency depth, dependency length, lexical difficulty, and word count. That approach differs from the present work in a basic way. Their control signal is feature-based and interpretable: the model is asked to rewrite a sentence to satisfy concrete structural constraints. In contrast, the present thesis uses grade-specific adapters and attempts to recover control indirectly through parameter composition. The advantage of the grade-based approach is that it maps more naturally onto educational categories such as school grade levels. The disadvantage is that it offers weaker control. The results in this thesis make that tradeoff fairly clear. Even the strongest fine-tuned adapters remain about one grade level away from their intended targets on average, and merging does not close that gap. Relative to a feature-controlled approach, this suggests that grade-level supervision alone may be too coarse to support precise readability control. The framework presented in Thillainathan and Koller (2024) also targets properties that are arguably closer to the underlying sources of processing difficulty than FKGL alone, even if those properties do not map cleanly onto standardized grade scales.

A second useful comparison is to controllable grade-specific simplification work such as Agrawal and Carpuat (2023). Their setup also treats grade level as a target variable, but they do not rely on grade labels alone. Instead, they predict low-level control values for the desired grade and use those values to guide generation. In effect, they decompose grade level into a group of more specific simplification operations. That is nearly the opposite of what this thesis does. Here, grade levels are encoded implicitly into separate adapters, and adapter merging attempts to interpolate between them. The present results suggest that this implicit approach captures relative differences between adjacent grade levels, but not with especially high precision. That makes the contrast with both Agrawal and Carpuat (2023) and Thillainathan and Koller (2024) particularly useful: their works suggest that finer-grained control variables may be necessary when the goal is not merely generic simplification, but simplification at a specific level.

The thesis also connects to work on adapters and parameter-efficient transfer more broadly, particularly recent work suggesting that independently trained adapters can be freely combined after training. Approaches such as AdapterSoup (Chronopoulou et al., 2023) emphasize the apparent modularity of adapter representations and suggest that multiple adapters can often

be mixed or averaged with little degradation in performance. The results in this thesis paint a more constrained picture. While grade-specific adapters do encode distinguishable simplification behaviors and can sometimes be recombined without catastrophic failure, the process is far less forgiving than the general modularity narrative might imply. Merge quality depends strongly on weight scaling, merge density, and the proximity of the merged adapters in grade space. Many merge configurations perform substantially worse than the underlying single-adapter baselines, and even the best mixtures do not improve readability control beyond what direct fine-tuning already achieves. Moreover, the anticipated interpolation effects from mixing adapters are relatively weak: combining neighboring grade adapters does not produce precise intermediate behavior, but rather coarse approximations that remain roughly one grade level away from the target. In this setting, adapter merging therefore functions less as a robust “mix-and-match” mechanism and more as a constrained form of parameter interpolation whose success depends on careful control of merge conditions.

Seen this way, the present work stands at an awkward but useful middle ground. It is in some ways simpler and more modular than explicit feature-controlled generation because new targets can be approximated by recombining existing adapters. But it is less precise than systems that control directly through interpretable linguistic features. That tradeoff may still be worthwhile in settings where grade-level categories matter more than exact structural constraints. At the same time, the results suggest that a stronger future system would probably combine the two perspectives rather than choose between them: modular adapters for efficiency, together with explicit feature-level control for precision.

7.6 Pedagogical Implications for SLA

One of the primary motivations for this work is the potential role of automatic TS in SLA environments. Instructors frequently adapt reading materials to match the proficiency level of their students, simplifying vocabulary, shortening sentences, or restructuring clauses while attempting to preserve the meaning and discourse structure of the original text. A system capable of reliably generating simplified text at multiple difficulty levels could therefore assist educators in producing accessible reading materials or adaptive learning resources.

The results of this study suggest that LLMs equipped with grade-specific adapters can capture broad differences in linguistic complexity that roughly correspond to educational grade levels. In practice, this means that the system can produce multiple variants of a sentence that differ in average lexical difficulty and syntactic complexity. From a pedagogical standpoint, this capability may be useful for generating approximate difficulty tiers of a text that can be used as starting points for instructional materials. At the same time, the experiments reveal clear limits on how precisely these models can control readability. Even the strongest fine-tuned adapters remain

approximately one grade level away from their intended targets on average, and adapter merging does not substantially improve that precision. For instructional settings, this level of variation may still be acceptable in most cases. Teachers rarely require reading materials to match a grade boundary exactly, and readability formulae themselves provide only approximate estimates of difficulty. However, the results indicate that current adapter-based approaches are better suited to coarse adjustments in difficulty rather than fine-grained control.

For these reasons, systems like the one explored in this thesis should be viewed primarily as tools to support educators. Automatic simplification could help generate candidate versions of a text at different approximate difficulty levels, allowing instructors to select or refine the versions that best match their pedagogical goals. In this role, the technology functions less as a fully automated simplification engine and more as a form of assistance for curriculum preparation.

7.7 Limitations

Several limitations of this study should be considered when interpreting the results.

First, the evaluation relies primarily on automated metrics such as FKGL, SARI, and BERTScore F1. These measures provide useful signals about lexical complexity, simplification quality, and semantic preservation, but they cannot fully capture how human readers experience simplified text. Readability formulae, in particular, are known to approximate difficulty using surface features such as sentence length and syllable count. While these features correlate with comprehension difficulty, they do not account for discourse structure, background knowledge, or pragmatic context. As a result, improvements in readability metrics do not necessarily translate directly into improved accessibility for language learners.

Second, the experiments operate within a simplified representation of readability based on discrete grade levels. This framework provides a convenient structure for experimentation, but linguistic complexity does not necessarily vary in clean steps between grade boundaries. The results suggest that grade-specific adapters encode overlapping simplification behavior rather than sharply separated grade representations. Consequently, the system can approximate nearby grade levels but struggles to achieve precise control over the specific metrics that determine grade level targets.

A third limitation concerns the merging methodology itself. Although the DARE_TIES merge operator proved stable under certain configurations, the experiments show that adapter merging is sensitive to weight scaling, merge density, and the relative proximity of the merged adapters. Many merge configurations lead to (often severely) degraded performance, and the stable operating conditions for merging appear relatively constrained. This suggests that the parameter

space of adapter composition may not yet be entirely understood, and that reliable merging may require additional constraints or improved merge operators.

Finally, the study focuses exclusively on sentence-level simplification. Many aspects of readability emerge across sentences, including discourse cohesion, narrative structure, and information flow. Extending the adapter-based approach to paragraph- or document-level simplification would likely introduce additional challenges related to maintaining coherence across longer passages.

CHAPTER 8

CONCLUSION

8.1 Summary of Findings

This thesis investigated whether modular adapter composition could provide precise control over readability levels in automatic text simplification. The central motivation was to explore whether adapters trained on discrete grade levels could be combined after training to generate text at specific and/or intermediate levels of linguistic complexity.

To evaluate this idea, we trained grade-specific simplification adapters using PEFT LoRA and examined whether their behavior could be combined through weighted parameter merging. The experiments explored several aspects of this approach, including merge window size, weight scaling strategies, adapter selection methods, and initialization strategies. In addition, dominance comparisons were used to evaluate whether certain weighting schemes or training strategies consistently outperformed others across grade levels.

The results show that grade-specific adapters do capture meaningful differences in linguistic complexity, but the resulting control over readability remains relatively coarse. Even the strongest fine-tuned adapters deviate from their target grade levels by approximately one grade on average. Adapter merging can approximate nearby grade levels and reproduce the behavior of individually trained adapters, but it does not improve accuracy beyond the strongest fine-tuned baselines. Moreover, many merge configurations lead to degraded performance when weight scaling or merge density moves outside a narrow stable region.

These findings suggest that adapter merging is better understood as a mechanism for compositional reuse of learned simplification behaviors rather than a method for achieving precise readability control. While merging allows intermediate or missing grade levels to be approximated without retraining new adapters, the interpolation effects are relatively weak and remain limited to local neighborhoods of adjacent grades.

Overall, the experiments answer the research questions of this thesis as follows. First, adapters trained on discrete grade levels do capture systematic differences in linguistic complexity, though the resulting control remains approximate rather than precise. Second, merging adapters does not improve readability accuracy beyond direct fine-tuning, but it enables flexible composition of grade representations. Third, weighted adapter mixtures can approximate nearby readability levels, although the results do not support the stronger hypothesis that merging provides continuous, dial-like control across the entire readability spectrum.

8.2 Contributions

This thesis makes several contributions to research on automatic text simplification, controllable generation, and parameter-efficient model composition.

At a high level, the work investigates whether adapter merging can be used as a mechanism for controlling linguistic complexity in NTS systems. By systematically evaluating a range of merging strategies and weighting schemes, the thesis clarifies both the potential and the limitations of using adapter composition for readability control.

The contributions can be grouped into theoretical, methodological, and practical categories.

Theoretical Contributions. This thesis contributes to our understanding of how linguistic complexity is represented within adapter-based simplification models.

First, the experiments demonstrate that grade-specific adapters encode overlapping simplification behaviors rather than sharply separated grade representations. Adjacent grade adapters often produce similar outputs, and merging them typically results in coarse interpolation rather than precise control. This finding suggests that readability levels learned from training data are largely overlapping and generalized simplifications rather than specific linguistic transformations.

Second, the results provide evidence that adapter merging does not inherently improve task performance relative to direct fine-tuning. While merged adapters can approximate the behavior of individually trained adapters, they rarely outperform them. This observation challenges optimistic assumptions in parts of the model-merging literature that independently trained modules can be freely recombined to produce improved behavior. Furthermore, the study shows that parameter merger does not necessarily correspond to smooth interpolation in linguistic output metrics. Even when merging adjacent grade adapters, the resulting readability control remains relatively coarse, indicating that the relationship between adapter parameters and textual complexity is more hazy than simple interpolation assumptions might suggest, based on the arithmetic nature of readability formulae.

Methodological Contributions. This thesis also contributes methodological insights into the use of adapter merging for controllable generation tasks.

The work presents a systematic evaluation of adapter merging for readability control, including experiments that vary merge window size, weight scaling, adapter selection strategies, and initialization methods. These experiments provide a detailed characterization of the conditions under which adapter merging remains stable. In particular, the results highlight the importance of maintaining normalized weight scaling and moderate merge density when combining adapters. Many alternative configurations produce unstable outputs or large deviations from the target grade level. These findings help define the practical constraints under which adapter merging can be applied reliably.

Practical Contributions for L2 Education. From an applied SLA perspective, this work explores how LLMs can support the creation of simplified reading materials for L2 learners.

The experiments demonstrate that adapter-based simplification models can generate multiple approximate levels of linguistic complexity from a single base model. This capability could assist educators in producing several simplified variants of a text without training separate models for each level.

At the same time, the results highlight the limitations of current readability control methods. The system can approximate grade levels but cannot reliably target precise grade boundaries. In practical educational settings, this suggests that automated simplification systems should be viewed as tools that assist instructors rather than fully automated course content generators. By clarifying both the potential and the limitations of adapter-based simplification, this thesis provides guidance for future research seeking to develop more precise and pedagogically useful methods for controllable text generation.

8.3 Future Work

The results of this thesis suggest several directions for future research.

One natural extension is to investigate alternative approaches to controlling linguistic complexity beyond grade-level supervision alone. The experiments indicate that discrete grade labels provide only coarse signals of readability. Future systems may benefit from incorporating more explicit linguistic control variables, as in Thillainathan and Koller (2024), such as syntactic depth, clause structure, or lexical frequency distributions. Combining adapter-based modularity with feature-level control mechanisms may provide more precise control over simplification behavior.

Future work should also evaluate these systems using human judgments rather than relying solely on automated metrics. While readability formulae and simplification metrics provide useful signals and effective measures for use during training, they cannot fully capture how language learners experience simplified text. Human evaluation would provide a more direct assessment of whether the generated outputs are genuinely easier to understand while preserving the meaning of the original content.

Finally, extending this approach beyond sentence-level simplification represents an important challenge. Many aspects of readability emerge across sentences and paragraphs, including discourse structure, coherence, and information flow. Investigating how adapter-based control mechanisms behave at the paragraph or document level could provide a more realistic picture of how these models might function in educational contexts.

Overall, the findings of this thesis suggest that adapter composition offers a promising foundation for modular simplification systems, but achieving precise and reliable readability control will require further refinement of both modeling and evaluation.

CHAPTER 9

POST SCRIPT - USE OF GENERATIVE AI TOOLS IN ACADEMIC RESEARCH

In accordance with the wishes of Dr. Prof. Alexander Koller in granting “[permission] to use those tools (generative digital assistants) without restrictions,” I am including this post-script to discuss the benefits and shortcomings of (current) generative AI tools and their uses in academic research and writing.

Throughout the development of this thesis, I used various AI tools for planning, coding, and writing assistance. These tools were at their best when I treated them as drafting aids for brainstorming structure, clarifying ideas, debugging, and reviewing the flow of the text. In the end, the experience was very mixed: the tools often acted as a second (patient) reviewer, but mostly they produced convincing sounding, but ultimately unhelpful suggestions that required careful consideration before incorporation.

Structural elements. ChatGPT was most helpful when planning the structure of chapters and sections. As I had never before written a master’s thesis, these suggestions often clarified what content typically belongs in the various chapters. When I was unsure where certain details should appear, it explained the purpose of each section and outlined what material to include and why it belongs in that section. In this sense, ChatGPT often functioned as a basic writing tutor. However, these structural suggestions generally require significant revision. It tended to over-generate organizational elements, proposing additional redundant sections that were unnecessary given the scope of this project. It also frequently produced outlines where multiple sections covered similar material. As a result, these suggestions were best viewed as loose guidelines, more specific than a generic “How to Write a Master’s Thesis in 42 Easy Steps” article, but as AI tools do not (yet) have access to my thoughts and intentions, it was essentially guessing at the kind of content and structure that would best fit this thesis.

Writing assistance. ChatGPT and Grammarly were only moderately useful during writing revision. In particular, they helped identify awkward transitions, gaps in explanations, and areas where the argumentation had become disjointed. Asking the model to explain the purpose of a section or paragraph often revealed weaknesses in the flow of the text. At the same time, these systems frequently produced repetitive or overly verbose suggestions. Suggested rewrites would add redundant explanations or introduce tangential ideas that were not supported by this research.

Reference checking. ChatGPT and turnitin performed poorly with regards to reference checking and assistance. When asked to identify claims that required citation or support, ChatGPT would produce entirely new paragraphs accompanied by fabricated references. Even when given a list of sources uploaded into the knowledge base for cross-referencing and checking for accidental plagiarism, the tools frequently suggested citations that were unrelated to the claim being made. It also showed an inexplicable penchant for citing “Mao et al. (2024)” as a reference to nearly every claim, and often created Franken-citations by mixing and matching authors, years, and portions of titles from actual related research. This is possibly the most disappointing shortcoming in using AI for help with academic writing, as maintaining a massive reference library of related information would be both incredibly useful, and in many ways, an obvious ideal use-case for AI systems. While AI writing tools can occasionally synthesize previous work to create logically cohesive argumentation, it cannot keep its own “thoughts” straight.

In an almost comical display of malfunctioning, turnitin consistently flagged even the most mundane content as missing citation: an early draft of this thesis was flagged over 1,500 times for ‘similar sources,’ including many stand-out examples, such as “Chapter 1. Introduction” (alongside most of the rest of the table of contents), nearly all glossary terms and definitions, direct in-text quotes with citations, as well as the entire bibliography of standardized entries.

Diagram generation. Another failed use-case was my attempt to use AI tools (e.g., ChatGPT, SORA) for generating diagrams or visual aids, which were by-and-large unsuccessful. While some models (GPT-5.x, Claude Sonnet-4.x) were capable of producing well-formatted LaTeX tables and complicated plots, they struggled with visually organized elements like flowcharts. Generated diagrams occasionally contained the correct components, but usually arranged them in overlapping layers that required tedious manual correction. Producing acceptable diagrams often required several iterations, and even then the results were inconsistent or ultimately unusable.

Debugging assistance. On the other hand, ChatGPT and Claude Sonnet proved particularly useful for debugging during technical development. When encountering implementation errors,

it was as simple as pasting the stack trace directly into the chat and describing the surrounding context. In many cases this approach proved faster and more effective than searching for the error message online and hoping to find a relevant forum post. Because the models could incorporate additional context from the chat log, I could iteratively narrow down the cause(s) of an issue. Another advantage was the ability of the model to synthesize potential solutions from a variety of sources. Rather than a single Stack Overflow post addressing my niche problem, the model could propose several plausible fixes derived from related discussions or technical documentation. Although the proposed solutions were not always correct, sometimes laughably bad, this process often accelerated the debugging cycle by quickly generating a list of specific testable suggestions and fixes to my code snippets.

Maintaining a critical eye. Overall, LLM tools proved most useful as scaffolding during the early stages of drafting and organization; suggesting structure, clarifying section purposes, and highlighting weaknesses in argumentation. At the same time, they consistently generate redundant or unreliable content. Therefore, we must treat AI-generated content as optimistic suggestions rather than reliable guidance. Every suggestion requires close reading before incorporation, and most require heavy editing. Used in this way, AI chat-bots function well as conversational brainstorming partners and writing tutors, even if they fall short in their specific suggestions. They provide a starting point and an additional perspective, but ultimately the responsibility to do the critical thinking remains squarely with the author.

APPENDIX A

FULL EXPERIMENTAL TABLES

Run Name	Δ FKGL	SARI	BERTScore F1
<i>Baseline Runs</i>			
off-the-shelf-baseline	3.528	33.619	87.045
graded-from-off-the-shelf	1.016	76.965	95.729
shared-warmup	1.184	75.878	95.922
graded-from-shared-warmup	1.034	76.498	95.867
dareties_g@all_ws@all_w@uniform_average	1.118	76.308	95.9
<i>Graded from off-the-shelf, Proximity Weighting Average @ 1</i>			
ots_dareties_ws@1_w@proximity_average	1.194	72.302	95.695
ots_dareties_ws@3_w@proximity_average	1.064	77.092	95.743
ots_dareties_ws@2_w@proximity_average	1.064	76.609	95.757
ots_dareties_ws@4_w@proximity_average	1.074	77.048	95.759
ots_dareties_ws@5_w@proximity_average	1.086	77.041	95.754
<i>Graded from off-the-shelf, Uniform Weighting Average @ 1</i>			
ots_dareties_ws@1_w@uniform_average	1.177	72.786	95.711
ots_dareties_ws@2_w@uniform_average	1.054	76.88	95.723
ots_dareties_ws@3_w@uniform_average	1.066	77.093	95.742
ots_dareties_ws@4_w@uniform_average	1.085	77.041	95.707
ots_dareties_ws@5_w@uniform_average	1.118	76.834	95.694

TABLE A.1: Performance of baseline models and off-the-shelf adapter merging configurations, averaged across target grades (2-12).

Run Name	Δ FKGL	SARI	BERTScore F1
<i>n-copies, Uniform weighting @ 1</i>			
dareties_ws@1-copies_w@uniform_average	1.035	76.43	95.875
dareties_ws@2-copies_w@uniform_average	1.415	66.918	95.636
dareties_ws@3-copies_w@uniform_average	1.106	75.16	95.715
dareties_ws@4-copies_w@uniform_average	1.055	75.991	95.766
dareties_ws@5-copies_w@uniform_average	1.044	76.377	95.842
dareties_ws@6-copies_w@uniform_average	1.027	76.548	95.848
dareties_ws@7-copies_w@uniform_average	1.035	76.512	95.855
dareties_ws@8-copies_w@uniform_average	1.032	76.525	95.853
dareties_ws@9-copies_w@uniform_average	1.031	76.607	95.859
dareties_ws@10-copies_w@uniform_average	1.037	76.461	95.865
dareties_ws@11-copies_w@uniform_average	1.036	76.533	95.856
<i>Uniform Weight Scaling ($w \cdot 0.5$)</i>			
dareties_ws@1_w@halved_average	2.589	40.325	90.928
dareties_ws@2_w@halved_average	2.131	48.171	93.547
dareties_ws@3_w@halved_average	1.977	51.654	94.361
dareties_ws@4_w@halved_average	1.89	53.319	94.706
dareties_ws@5_w@halved_average	1.862	53.882	94.82
<i>Uniform Weight Scaling ($w \cdot 1.0$)</i>			
dareties_ws@1_w@uniform_average	1.379	71.067	95.096
dareties_ws@2_w@uniform_average	1.076	76.079	95.821
dareties_ws@3_w@uniform_average	1.056	76.346	95.845
dareties_ws@4_w@uniform_average	1.065	76.435	95.872
dareties_ws@5_w@uniform_average	1.086	76.454	95.875
<i>Uniform Weight Scaling ($w \cdot 2.0$)</i>			
dareties_ws@1_w@doubled_average	2.294	73.054	91.46
dareties_ws@2_w@doubled_average	12.541	47.056	48.132
dareties_ws@3_w@doubled_average	16.356	37.158	32.03
dareties_ws@4_w@doubled_average	18.441	31.279	22.06
dareties_ws@5_w@doubled_average	19.485	28.036	16.611
<i>Uniform Weight Scaling ($w \cdot 3.0$)</i>			
dareties_ws@1_w@tripled_average	18.839	29.516	23.054
dareties_ws@2_w@tripled_average	24.057	35.894	49.685
dareties_ws@3_w@tripled_average	32.736	35.052	53.437
dareties_ws@4_w@tripled_average	33.628	35.764	56.237
dareties_ws@5_w@tripled_average	29.818	38.225	61.117

TABLE A.2: Performance of n-copies and uniform scaling adapter merging configurations, averaged across target grades (2-12).

Run Name	Δ FKGL	SARI	BERTScore F1
<i>Proximity Weight Scaling ($w \cdot \frac{d}{x^{\frac{1}{2}}}$)</i>			
dareties_ws@1_w@proximity-flatter_average	1.189	72.933	95.667
dareties_ws@2_w@proximity-flatter_average	1.072	75.902	95.785
dareties_ws@3_w@proximity-flatter_average	1.062	76.31	95.842
dareties_ws@4_w@proximity-flatter_average	1.059	76.501	95.848
dareties_ws@5_w@proximity-flatter_average	1.071	76.409	95.875
<i>Proximity Weight Scaling ($w \cdot \frac{d}{x^1}$)</i>			
dareties_ws@1_w@proximity_average	1.19	72.892	95.662
dareties_ws@2_w@proximity_average	1.083	75.813	95.773
dareties_ws@3_w@proximity_average	1.067	76.236	95.784
dareties_ws@4_w@proximity_average	1.067	76.192	95.806
dareties_ws@5_w@proximity_average	1.076	76.205	95.802
<i>Proximity Weight Scaling ($w \cdot \frac{d}{x^2}$)</i>			
dareties_ws@1_w@proximity-squared_average	1.264	70.963	95.613
dareties_ws@2_w@proximity-squared_average	1.153	74.58	95.693
dareties_ws@3_w@proximity-squared_average	1.148	74.853	95.683
dareties_ws@4_w@proximity-squared_average	1.171	74.484	95.692
dareties_ws@5_w@proximity-squared_average	1.211	73.657	95.695
<i>Proximity Weight Scaling ($w \cdot \frac{d}{x^3}$)</i>			
dareties_ws@1_w@proximity-cubed_average	1.419	67.066	95.539
dareties_ws@2_w@proximity-cubed_average	1.332	69.158	95.67
dareties_ws@3_w@proximity-cubed_average	1.432	66.299	95.683
dareties_ws@4_w@proximity-cubed_average	1.582	61.762	95.472
dareties_ws@5_w@proximity-cubed_average	1.753	56.731	95.077
<i>Random Weights, Average @ 1</i>			
dareties_trial@1_w@random-1_average	1.152	75.131	95.694
dareties_trial@2_w@random-1_average	1.163	75.341	95.726
dareties_trial@3_w@random-1_average	1.195	74.372	95.741
<i>Random Weights, Average @ 2</i>			
dareties_trial@1_w@random-2_average	4.643	67.023	83.394
dareties_trial@2_w@random-2_average	5.634	64.521	79.044
dareties_trial@3_w@random-2_average	7.615	59.241	70.388
<i>Random Weights, Average @ 3</i>			
dareties_trial@1_w@random-3_average	20.044	25.263	15.485
dareties_trial@2_w@random-3_average	20.794	24.124	13.495
dareties_trial@3_w@random-3_average	19.864	26.566	18.385

TABLE A.3: Performance of proximity steepness and random weight adapter merging configurations, averaged across target grades (2-12).

Run Name	Δ FKGL	SARI	BERTScore F1
<i>Adapter Selection, Downshifted 1 Grade</i>			
dareties_ws@1_w@uniform_downshifted-1	1.173	73.962	95.665
dareties_ws@2_w@uniform_downshifted-1	1.108	74.313	95.704
dareties_ws@3_w@uniform_downshifted-1	1.073	75.924	95.793
dareties_ws@4_w@uniform_downshifted-1	1.059	76.254	95.829
dareties_ws@5_w@uniform_downshifted-1	1.076	76.373	95.851
<i>Adapter Selection, Downshifted 2 Grades</i>			
dareties_ws@1_w@uniform_downshifted-2	1.204	74.024	95.66
dareties_ws@2_w@uniform_downshifted-2	1.127	74.619	95.706
dareties_ws@3_w@uniform_downshifted-2	1.107	74.399	95.674
dareties_ws@4_w@uniform_downshifted-2	1.079	75.921	95.773
dareties_ws@5_w@uniform_downshifted-2	1.073	76.273	95.813
<i>Adapter Selection, Upshifted 1 Grade</i>			
dareties_ws@1_w@uniform_upshifted-1	1.122	74.838	95.735
dareties_ws@2_w@uniform_upshifted-1	1.099	75.688	95.875
dareties_ws@3_w@uniform_upshifted-1	1.064	76.444	95.896
dareties_ws@4_w@uniform_upshifted-1	1.126	76.537	95.895
dareties_ws@5_w@uniform_upshifted-1	1.078	76.486	95.909
<i>Adapter Selection, Upshifted 2 Grades</i>			
dareties_ws@1_w@uniform_upshifted-2	1.114	74.566	95.733
dareties_ws@2_w@uniform_upshifted-2	1.092	75.582	95.884
dareties_ws@3_w@uniform_upshifted-2	1.17	75.879	95.951
dareties_ws@4_w@uniform_upshifted-2	1.082	76.473	95.93
dareties_ws@5_w@uniform_upshifted-2	1.141	76.401	95.943
<i>Adapter Selection, No Target Grade</i>			
dareties_ws@1_w@uniform_no-tgt-grade	1.351	68.964	95.763
dareties_ws@2_w@uniform_no-tgt-grade	1.162	73.635	95.695
dareties_ws@3_w@uniform_no-tgt-grade	1.092	75.911	95.832
dareties_ws@4_w@uniform_no-tgt-grade	1.074	76.201	95.856
dareties_ws@5_w@uniform_no-tgt-grade	1.09	76.207	95.888

TABLE A.4: Performance of adapter selection merging configurations, averaged across target grades (2-12).

Run Name	Δ FKGL	SARI	BERTScore F1
<i>Proximity Weighting, Sum to 1</i>			
dareties_ws@1_w@proximity_sum	3.128	35.291	88.446
dareties_ws@2_w@proximity_sum	3.293	34.614	87.91
dareties_ws@3_w@proximity_sum	3.407	34.334	87.625
dareties_ws@4_w@proximity_sum	3.457	34.201	87.455
dareties_ws@5_w@proximity_sum	3.466	34.048	87.372
<i>Uniform Weighting, Sum to 1</i>			
dareties_ws@1_w@uniform_sum	3.084	35.436	88.533
dareties_ws@2_w@uniform_sum	3.268	34.652	87.983
dareties_ws@3_w@uniform_sum	3.375	34.334	87.692
dareties_ws@4_w@uniform_sum	3.418	34.219	87.513
dareties_ws@5_w@uniform_sum	3.469	34.163	87.413
<i>Linear Merge, Proximity Weighting, Average to 1</i>			
linear_ws@1_w@proximity_average	111.147	20.26	68.24
linear_ws@2_w@proximity_average	25.038	17.70	63.93
linear_ws@3_w@proximity_average	22.70	17.69	63.64
linear_ws@4_w@proximity_average	22.70	17.68	63.28
linear_ws@5_w@proximity_average	22.70	17.69	63.79
<i>Linear Merge, Proximity Weighting, Sum to 1</i>			
linear_ws@1_w@proximity_sum	30.386	37.73	56.41
linear_ws@2_w@proximity_sum	93.276	24.12	64.61
linear_ws@3_w@proximity_sum	112.358	22.16	71.07
linear_ws@4_w@proximity_sum	81.580	19.20	69.90
linear_ws@5_w@proximity_sum	61.196	18.17	68.67
<i>Linear Merge, Uniform Weighting, Average to 1</i>			
linear_ws@1_w@uniform_average	109.529	19.91	67.95
linear_ws@2_w@uniform_average	22.855	17.70	63.98
linear_ws@3_w@uniform_average	22.70	17.68	63.28
linear_ws@4_w@uniform_average	22.70	17.68	63.28
linear_ws@5_w@uniform_average	22.70	17.68	63.28
<i>Linear Merge, Uniform Weighting, Sum to 1</i>			
linear_ws@1_w@uniform_sum	32.466	36.59	57.52
linear_ws@2_w@uniform_sum	98.037	22.61	64.18
linear_ws@3_w@uniform_sum	101.129	20.95	70.34
linear_ws@4_w@uniform_sum	65.783	18.54	69.13
linear_ws@5_w@uniform_sum	48.456	17.81	67.80

TABLE A.5: Performance of linear adapter merging and dare_ties sum configurations, averaged across target grades (2-12).

dFKGL by window_size (mean + median, averaged across grades) (all runs) [page 1]

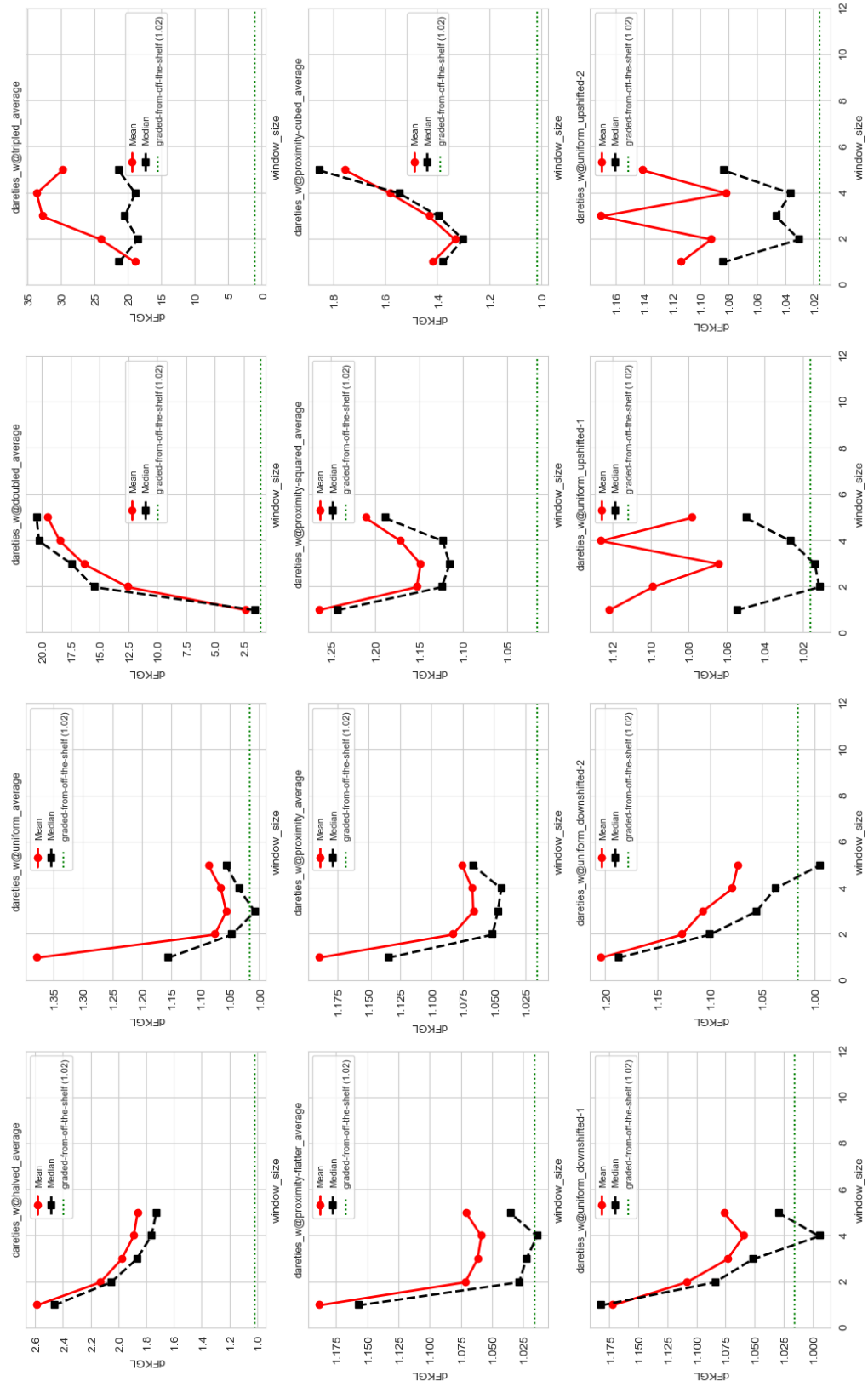


FIGURE A.1: Window size results (a).

dFKGL by window_size (mean + median, averaged across grades) (all runs) [page 2]

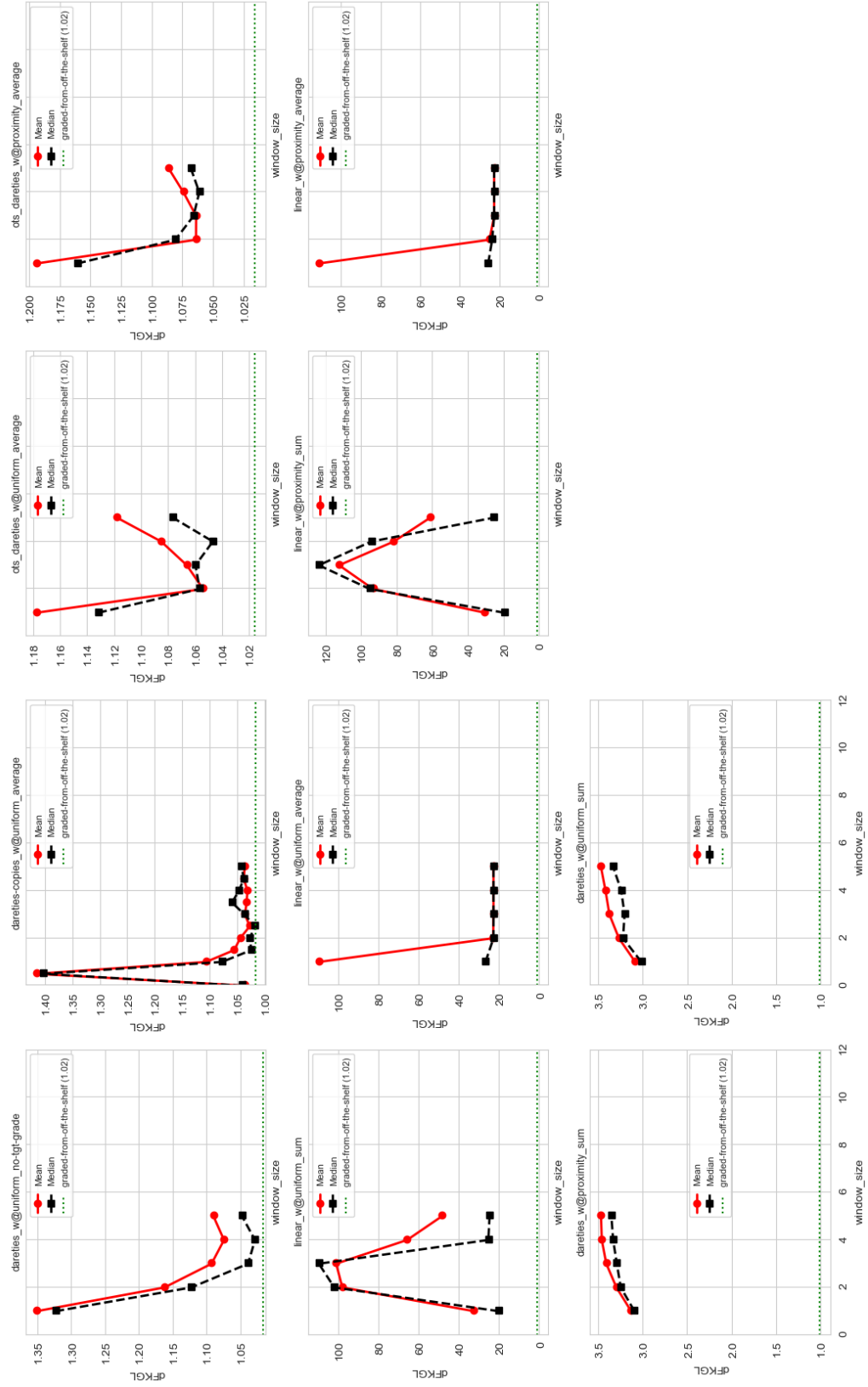


FIGURE A.1: Window size results (b).

n_adapters vs dFKGL by weight_scheme (all runs) [page 1]

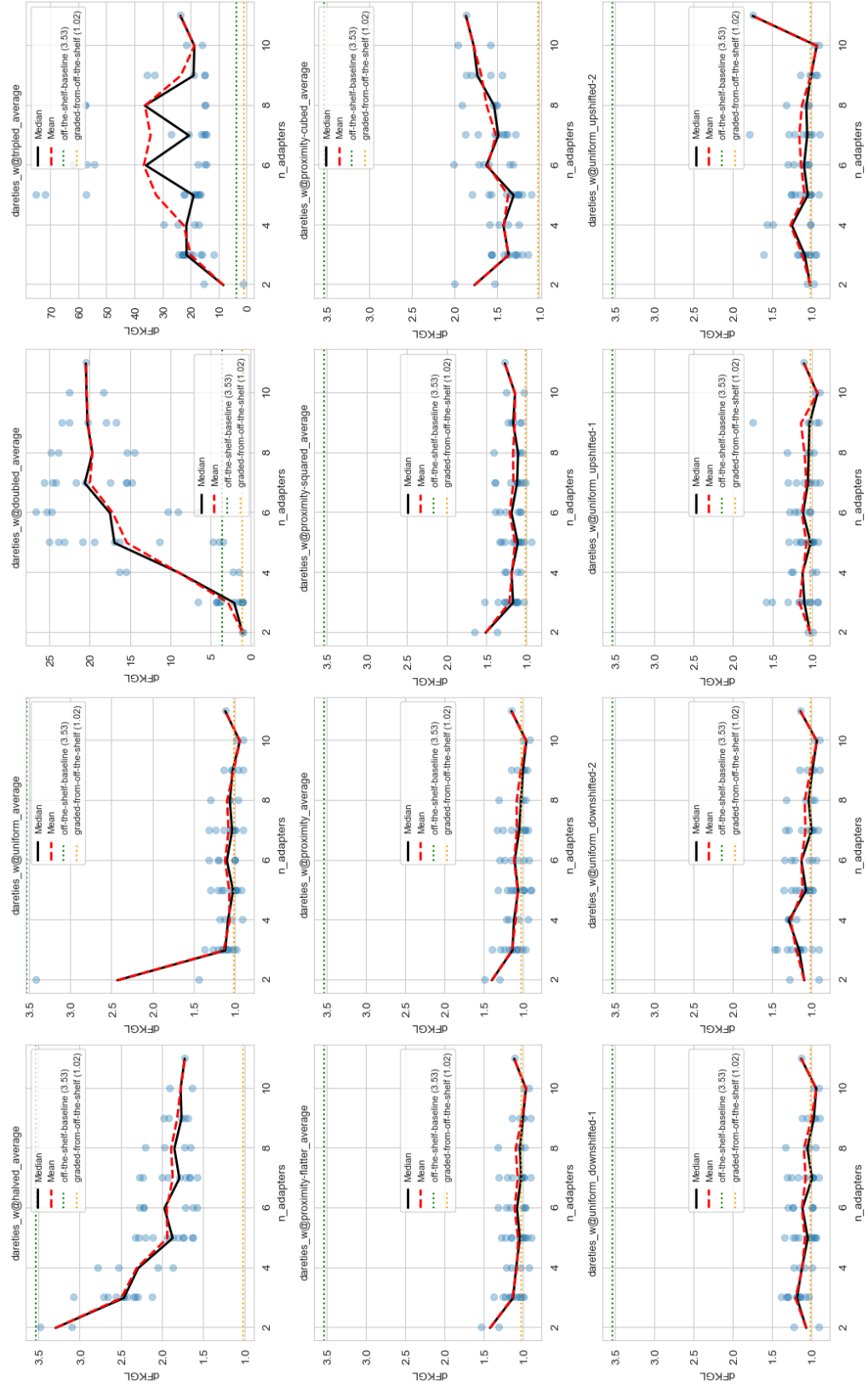


FIGURE A.2: Number of adapters results (a).

n_adapters vs dFGL by weight_scheme (all runs) [page 2]

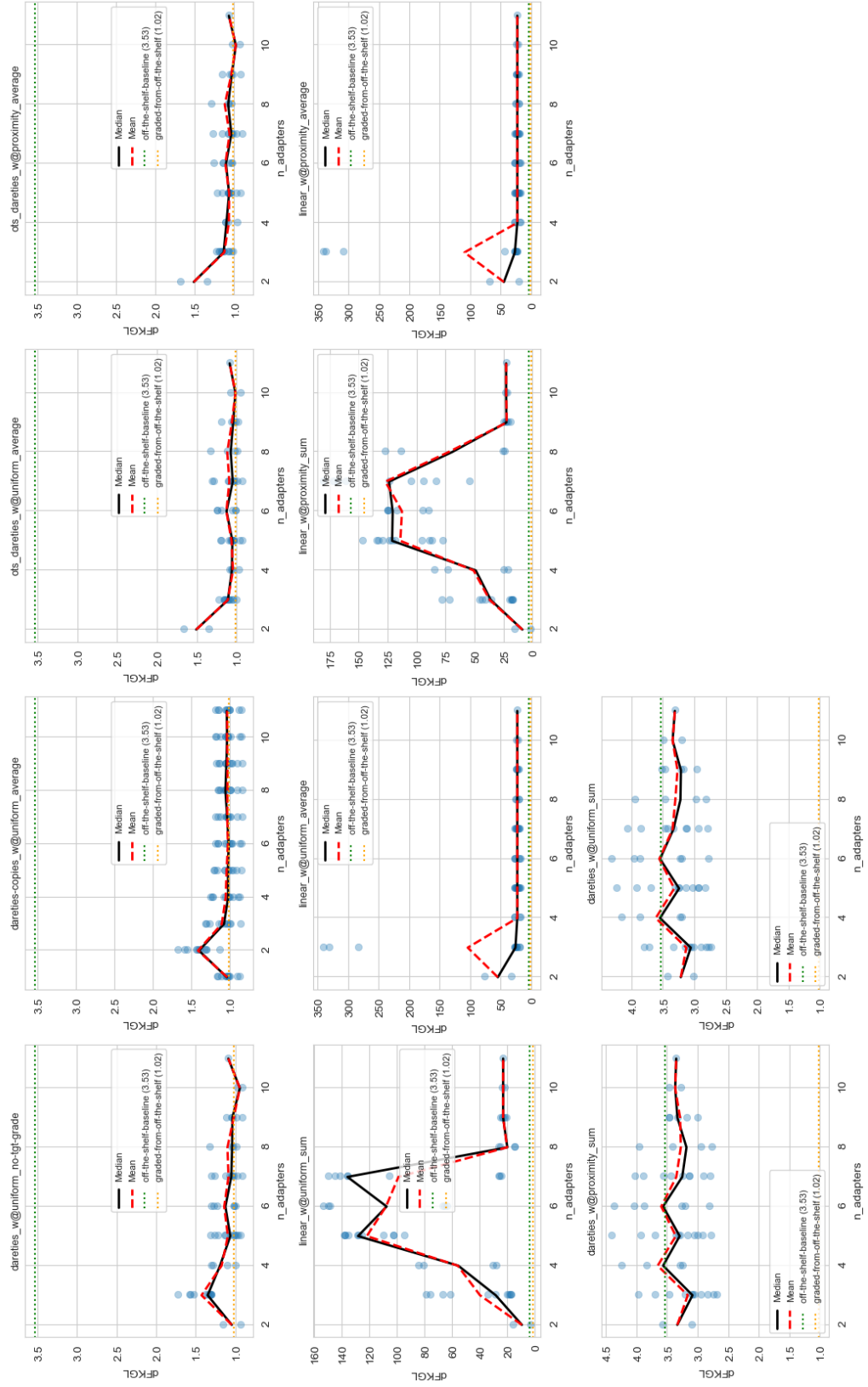


FIGURE A.2: Number of adapters results (b).

APPENDIX B

MODEL INPUT-OUTPUT EXAMPLES

Success Case - Shared Warm-up Adapter

Prompt: user

Rewrite this Input sentence to make it easily understandable by students in Grade 5

Atticus is a lawyer and resident of the fictional Maycomb County, Alabama, and the father of Jeremy Atticus “Jem” Finch and Jean Louise “Scout” Finch.

assistant

Source: user

Rewrite this Input sentence to make it easily understandable by students in Grade 5

Atticus is a lawyer and resident of the fictional Maycomb County, Alabama, and the father of Jeremy Atticus “Jem” Finch and Jean Louise “Scout” Finch.

Input: {'prompt_token_ids': [128000, 128006, 882, 128007, 271, 59565, 1269, 420, 5688, 11914, 311, 1304, 433, 6847, 49839, 555, 4236, 304, 24749, 220, 20, 198, 10673, 59344, 374, 264, 15779, 323, 19504, 315, 279, 44682, 3297, 22631, 6406, 11, 21272, 11, 323, 279, 7126, 315, 28217, 7867, 59344, 330, 41, 336, 1, 96333, 323, 20263, 56578, 330, 3407, 412, 1, 96333, 13, 128009, 128006, 78191, 128007, 271], 'multi_modal_data': None}

Pred: Atticus is a lawyer and lives in Maycomb County, Alabama. He is the father of Jeremy Atticus “Jem” Finch and Jean Louise “Scout” Finch.

Label: Atticus is a lawyer who lives in Alabama. He is the father of Jeremy Atticus “Jem” Finch and Jean Louise “Scout” Finch.

Edge case - Shared Warm-up Adapter

Prompt: user Rewrite this Input sentence to make it easily understandable by students in Grade 7

Fourteen players have served as the captain of the Calgary Flames, beginning with Brad Marsh in 1980-81.

assistant

Source:

Fourteen players have served as the captain of the Calgary Flames, beginning with Brad Marsh in 1980-81.

Input: {'prompt_token_ids': [128000, 128006, 882, 128007, 271, 59565, 1269, 420, 5688, 11914, 311, 1304, 433, 6847, 49839, 555, 4236, 304, 24749, 220, 22, 198, 28070, 15247, 4311, 617, 10434, 439, 279, 24862, 315, 279, 38975, 73297, 11, 7314, 449, 17478, 40163, 304, 220, 3753, 15, 29005, 330, 5932, 13, 128009, 128006, 78191, 128007, 271], 'multi_modal_data': None}

Pred: Fourteen players have served as the captain of the Calgary Flames. The first captain was Brad Marsh in 1980-81.

Label: Fourteen players have been the captain of the Calgary Flames, beginning with Brad Marsh in 1980-81.

Failure Case - Magnitude prune:

Prompt: user

Rewrite this Input sentence to make it easily understandable by students in Grade 2

He married his wife Adrienne in 1972.

assistant

Source: user

Rewrite this Input sentence to make it easily understandable by students in Grade 2

He married his wife Adrienne in 1972.

```
Input: {'prompt_token_ids': [128000, 128006, 882, 128007, 271, 59565, 1269, 420, 5688, 11914, 311, 1304, 433, 6847, 49839, 555, 4236, 304, 24749, 220, 17, 198, 1548, 12502, 813, 7555, 62443, 26193, 304, 220, 4468, 17, 13, 128009, 128006, 78191, 128007, 271], 'multi_modal_data': None}
```

[illegible]

Label: He married his wife, Adrienne, in 1972.

APPENDIX C

DATA CLEANING CLASSIFIER

C.0.1 Feature Definitions.

Each source-target sentence pair is represented using three scalar features designed to capture (in)appropriate simplification behavior.

Semantic Similarity. Source and target sentences are encoded using spaCy’s `en_core_web_sm` `tok2vec` embeddings (Honnibal and Montani, 2015). Cosine similarity between the two sentence embeddings is used as a proxy for semantic equivalence:

$$\text{sim}(s, t) = \frac{s \cdot t}{\|s\| \|t\|}$$

Values range from -1 to 1 , where higher values indicate greater semantic overlap.

Length Ratio. Length ratio measures the relative size of the simplified sentence:

$$\text{length ratio} = \frac{\text{words}_{\text{target}}}{\text{words}_{\text{source}}}$$

Values below 1 indicate shorter simplified outputs, while extremely small ratios often correspond to missing or incomplete simplifications.

Grade Difference. Grade difference measures the change in readability between the source and target sentences using ΔFKGL :

$$\Delta\text{FKGL} = \text{FKGL}_{\text{target}} - \text{FKGL}_{\text{source}}$$

Negative values indicate a reduction in readability grade level, which is consistent with successful simplification.

C.0.2 Labeling Protocol

A random sample of 1,000 sentence pairs was manually inspected to construct training labels for the classifier. Each pair was assigned one of two labels:

- **Valid simplification (1):** the target sentence preserves the meaning of the source sentence while reducing linguistic complexity.
- **Invalid pair (0):** the target sentence is missing, unrelated, semantically inconsistent with the source sentence, or does not meaningfully simplify the original text.

Annotation was performed by the author through manual inspection of each pair. Examples containing empty targets, extremely short outputs, or unrelated content were consistently labeled as invalid.

Figure C.1 visualizes the manually labeled sample in feature space, illustrating that valid and invalid simplifications form separable clusters along these dimensions.

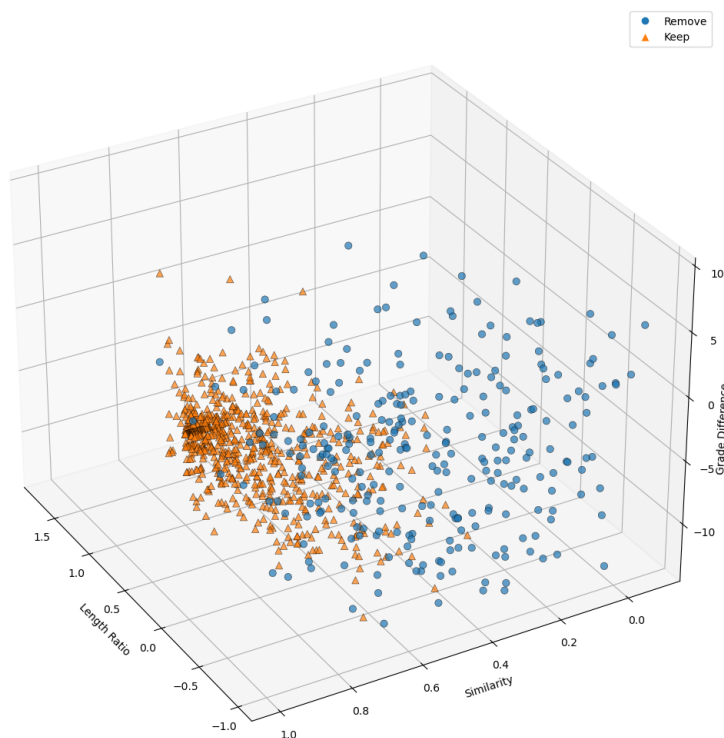


FIGURE C.1: Manually labeled simplification pairs plotted using the three filtering features.

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